

SYSTEMATIC TRADING MTH9897

Lecture 1: Fixed Income

Quantitative Investment Framework



Misha Boroditsky, PhD
Fixed Income Analytics Lead
Bloomberg LLP
mboroditsky@hotmail.com

Dmitry Rakhlin, PhD
Managing Director, Portfolio Management
WorldQuant LLC
Dmitry.Rakhlin@gmail.com

Teaching Assistant:
Alan Coman alancoman@gmail.com

COURSE STRUCTURE

Scope: Overview Of Popular Themes in Quant Trading

Lectures 1-2 (Sep 1, September 8) Systematic Fixed Income by Misha Boroditsky

Lectures 3,4,5 (September 15, 22, 29) Fundamental Quant Models by Dmitry Rakhlin

- Part 1: Common market anomalies. Behavioral Finance. Fama-French 3 & 5 factor models
- Part 2: More anomalies. “Factor zoo”.
- Part 3: Factor-mimicking portfolios. Portfolio construction.
- Part 4: Factor diagnostics. The role of factor decay. Cost of strategy implementation.

Lecture 6 (October 6) - invited lecture on capital markets by Peter Sheridan, Managing Director UBS.

Lecture 7 (October 13)

- Fixed Income factor models by Misha Boroditsky (1 hour)
- Final project review/consultations (the rest)

Attendance We will need you to fill out signup sheets to track attendance

There will be 3 HWs !!! (due 9/15, 10/6 and 10/20) **individual or in pairs**

- Fixed Income
- Equities
- Final Exam (October 20th) - Course project presentation

General Resources (favorite books)

- [The Complete Guide to Capital Markets for Quantitative Professionals](#) by Alex Kuznetsov
- [Arbitrage Theory in Continuous Time](#) by Thomas Bjork
- [Theory of Financial Risk and Derivative Pricing: From Statistical Physics to Risk Management](#) by JP Bouchaud
- [Quantitative Portfolio Management: The Art and Science of Statistical Arbitrage](#) by Michael Isichenko
- [The Mythical Man-Month: Essays on Software Engineering](#), by Frederick Brooks Jr

- ☐ Research, replicate and (hopefully) improve existing investment strategy
- ☐ Working individually or in groups of 2 – all three homeworks
- ☐ You may use any framework (R, Matlab, Python, Java ...). The code will be reviewed
- ☐ A list of suggested strategies will be given. Examples:
 - Value/Growth/Quality quant models
 - Market Sentiment / Sector Rotation model
- ☐ Feel free to suggest your own strategy.
 - Pre-requisite: there should be a publication/white paper/research paper
 - Instructors will need to review it before approving

Brief History of Bond Market

*“Hence the extraordinary growth of a class, or rather, of a stratum of rentiers, i.e., people who live by '**clipping coupons**', who take no part in any enterprise whatever, whose profession is idleness.”*

Vladimir Lenin, "[Imperialism, the Highest Stage of Capitalism](#)", 1920

- In the early 1600s, the Dutch East India Trading Co. was the first private company to offer corporate bonds to the public
- The first-ever Sovereign bond was issued in 1693 by the newly formed Bank of England.
- 1970's Rise of Mortgage-backed securities
- 1990's Futures, junk bonds, CMO's, Municipals
- 1994 Credit Default Swaps (Bankers' Trust, JPM)
- 2010's Fixed Income ETF's
- 2015 electronic credit trading
- 2020 Portfolio Trading



Outline

- Intro
 - FI vs Equities
 - Buy Side vs Sell side
- Fixed Income Primer
- Corporate Bonds
- Fixed Income ETFs and portfolio trading
- Municipal Bonds
- Trading System for buy-side vs sell-side
- Systematic factor Investing in credit

“What is a bond” <https://www.youtube.com/watch?v=FKPPeEA43po>

Fixed Income Vs Equities Trivia

Measure	Fixed Income (Corp)	Equities
Market Size,\$	11.5T	47T
Issuance,\$	2.3T	400B
Daily Trading Volume,\$	46B	481B
Electronic Volume,\$	3B (*)	350B
Number of trades	70k	59m
Electronic Trades	12.5k	50m
Instruments	100K	10K
Average trade size,\$	650K	8K
Electronic trade size,\$	240K	7K

*Sources: FINRA, SIFMA 2025

Why Organizations Trade

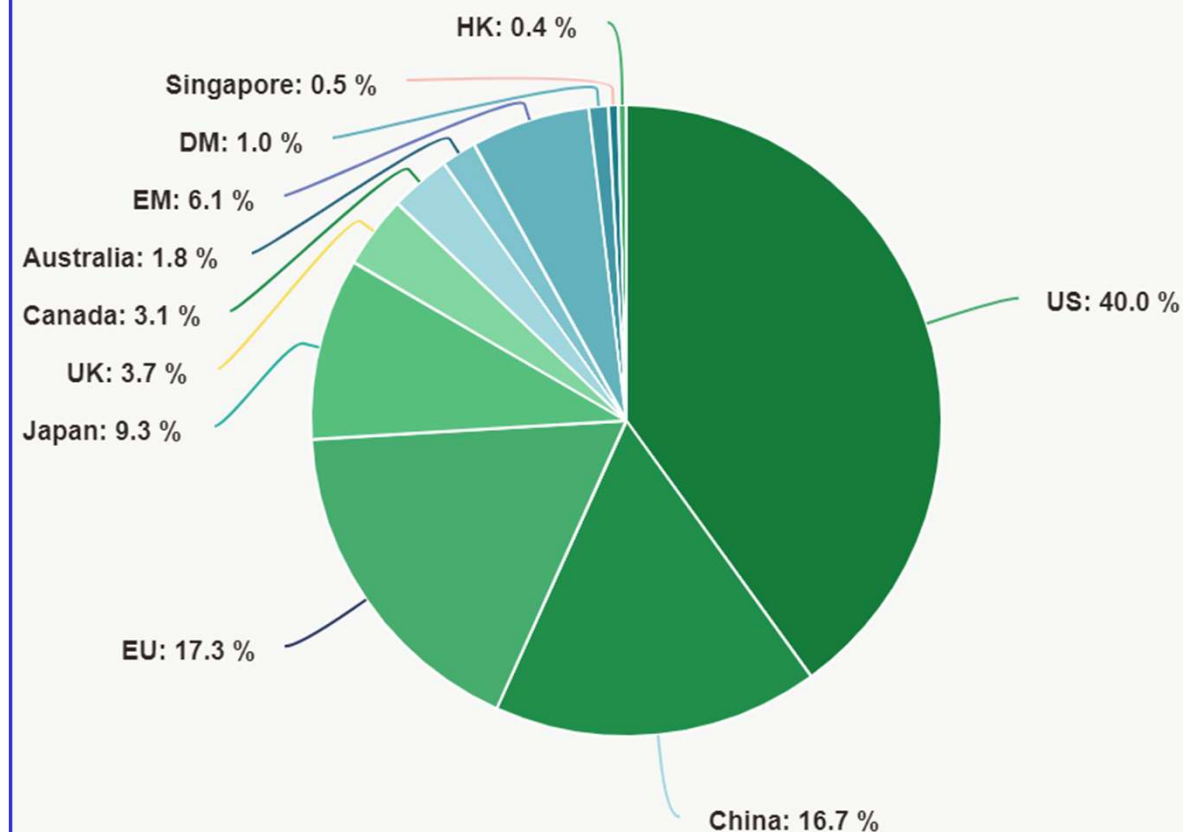
SELL-SIDE	BUY-SIDE
- Provides service (makes markets) in competition - Service provider to buy-side - Market centers' customer - Revenue $\sim$ BidAskSpread $\times$ Vol - Spread ~ 0.1bp for OTR bonds - Spread ~ 10-50bp for corporate bonds and munis Brokerage pie: \$640B (US only)	- Provides service to end clients (real money) in competition - Is a customer to sell-side - Market centers' customer - Revenue $\sim$ fee $\times$ AUM fees 10-60 bp (mm, fi, equity) $\text{AUM}(t) = f(\text{sales}(t-1y), \text{performance}(t-4y), \text{marketCond}(t))$ AssetMgmt pie \$550B on \$136T

<https://www.sifma.org/resources/research/fact-book/>

Global Fixed Income Market

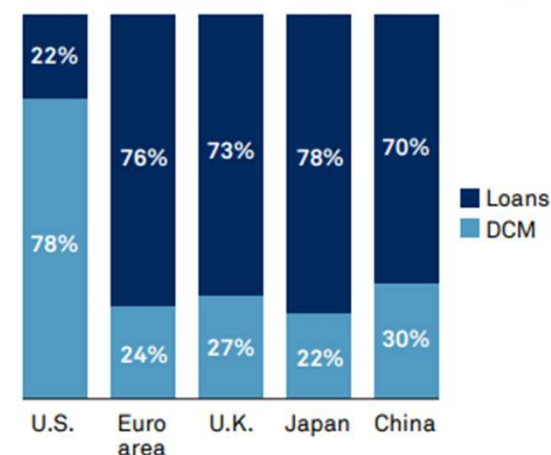
Global Fixed Income Outstanding

Source: Bank for International Settlements, SIFMA estimates



\$130 Trillion market
Comparable to equities
Linked instruments
Huge liquidity spectrum
Bonds are not the only way to borrow

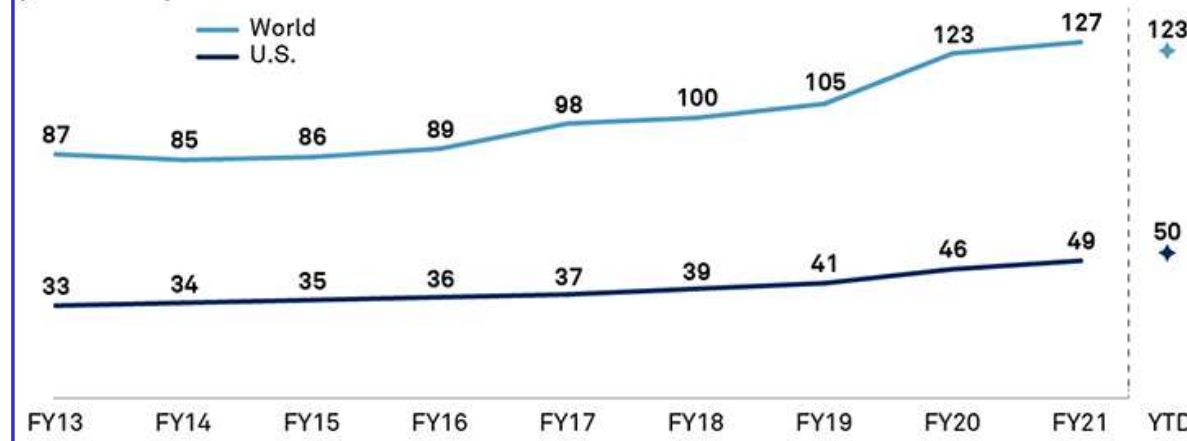
Debt Financing of Non-Financial Corporations



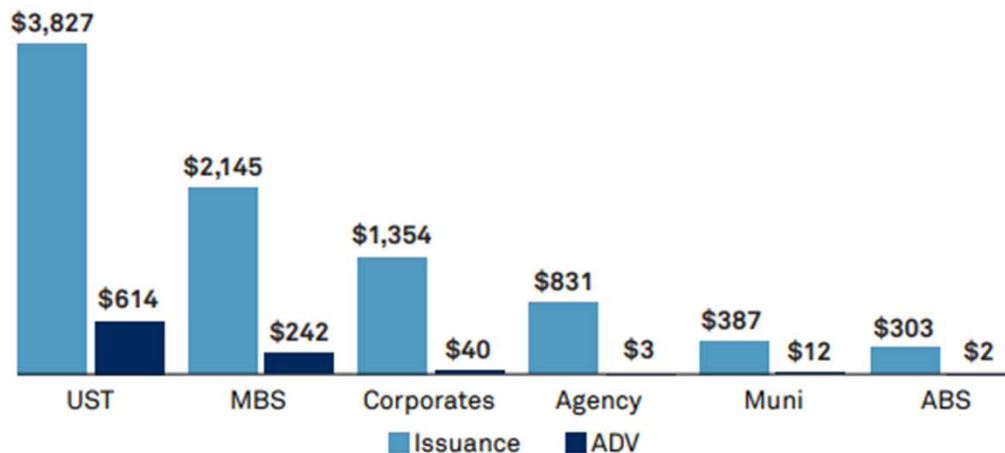
Note: As of 2021, China 2019; DCM = debt capital markets = corporate bonds only.
Source: U.S. Federal Reserve, ECB, Bank of Japan, National Bureau of Statistics of China

Fixed Income Market Size

U.S. Share of Global Fixed-Income Markets (\$ Trillion)



Market Size by Asset Class (Outstanding/Issuance) (\$ Billion)

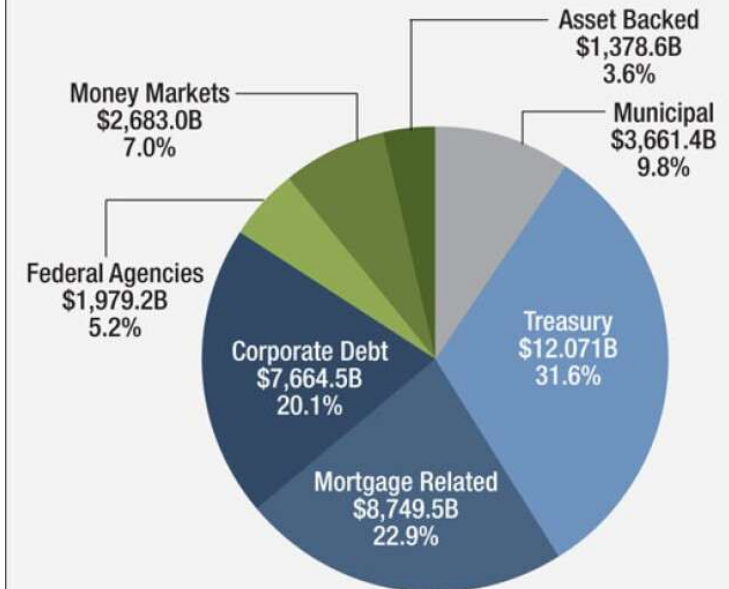


Source: SIFMA

<https://www.sifma.org/wp-content/uploads/2023/05/Understanding-Fixed-Income-Markets-2023-23-2007-FINAL.pdf>

Bond Market Breakdown

Fixed income debt in trillions of dollars



Source: SIFMA

Note how much ADV to issuance ratio varies: some instruments get parked faster, e.g. to pension funds and insurance accounts

Rates vs Credit

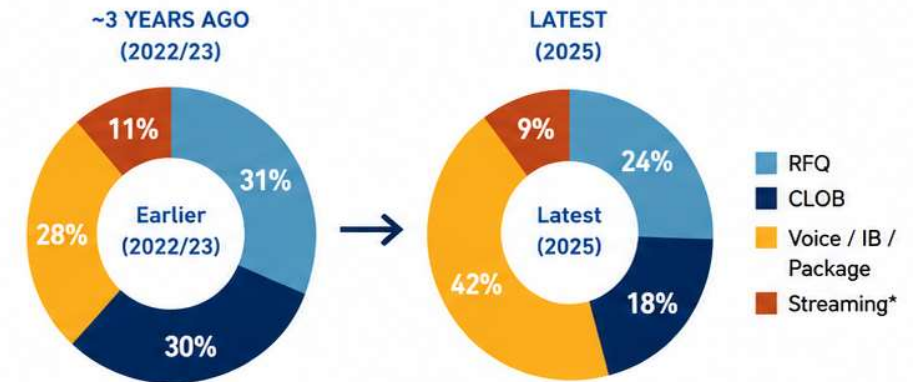
U.S. TREASURIES ELECTRONIC TRADING

By Platform (notional share)



Note: Platform shares for 2024/25 are rounded. Source: Coalition Greenwich

By Channel

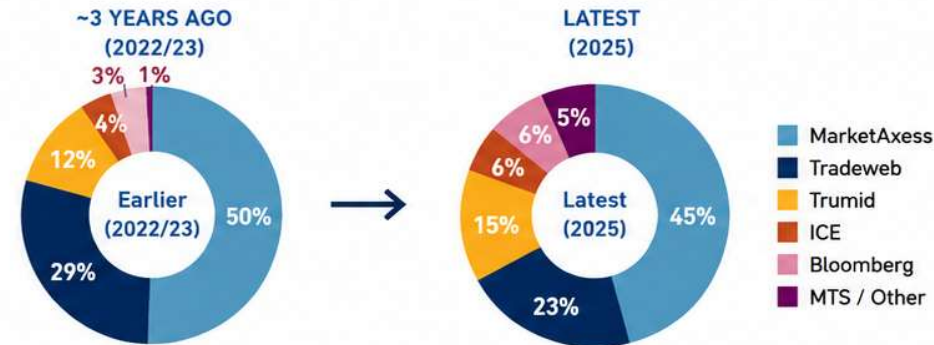


Note: 2025 reflects Jan–Dec 2025 average shares.

*Streaming often embedded in RFQ/CLOB workflows. Source: Coalition Greenwich

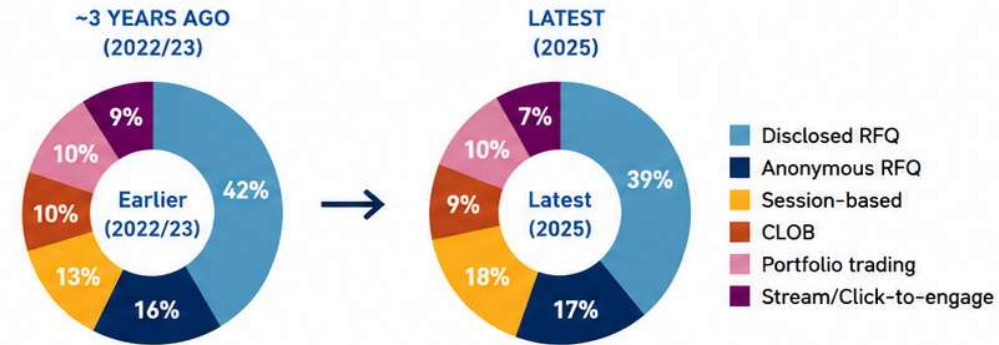
U.S. CORPORATE BONDS ELECTRONIC TRADING

By Platform (notional share)



Source: Coalition Greenwich

By Channel



Source: Coalition Greenwich

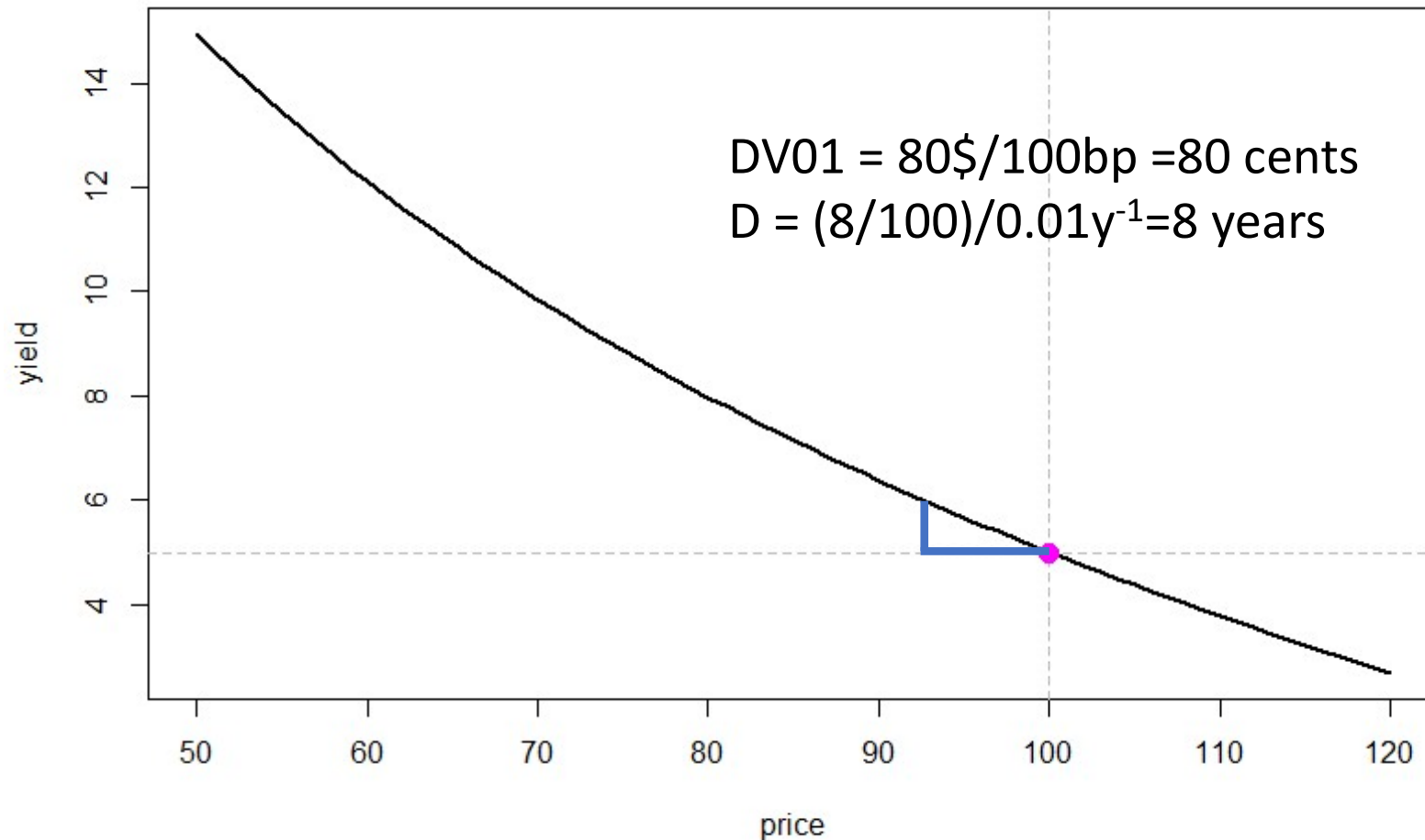
Percentages may not sum to 100 due to rounding. Source: Coalition Greenwich MarketView (2022/23, 2024/25, 2025).

Bond primer

- “Atomic” instrument is a zero-coupon bond a.k.a. bill
 - You pay price P get face value F back at time T
 - Yield y is rate of return: $P = F * \exp(-yT) \equiv Fd$, where d is the discount factor
 - For a given bond, yield defines price, and vice versa
 - yield and price move in opposite directions
 - Who decides what y is today? Is it known for every time T ?
 - Typically, zero-coupon bonds are issued up to 1 year (and called bills).
- Bonds ($T=30$ years) and Notes (in-between) pay coupons, typically twice a year, so $n=2T$
 - $P = \sum_{k=1}^{2n} \frac{C_k/2}{(1+y/2)^k} + \frac{F}{(1+y/2)^{2n}} \equiv \sum_{k=1}^n C_k d_k + Fd_n$
 - C_k – coupon (may be different), F is face value, y is yield, d discount factor
 - Nothing to do with bond pricing. These are just two ways to describe bond value.
 - A bond can be quoted on price (percentage), yield, or discount.
 - Note that $y \rightarrow P$ is easier than $P \rightarrow y$
- $DV01 = -dP/dy * 1bp = -0.0001 * dP/dy$
 - Dollar Value of a basis points
 - Measures bond – or bond portfolio – value’s sensitivity to change in yields
- Duration
 - $dP/P = d(\log P) \equiv -D dy$
 - has dimension of years
 - Describes sensitivity to interest rates – used in risk management
 - Applies to an instrument of a portfolio, fund, or ETF
 - $D = 10000 * DV01 / P$
 - For zero coupon bond $D = T$; coupon bonds $D < T$ as can be seen from DCF equation

“Bond prices are up... Yields down”

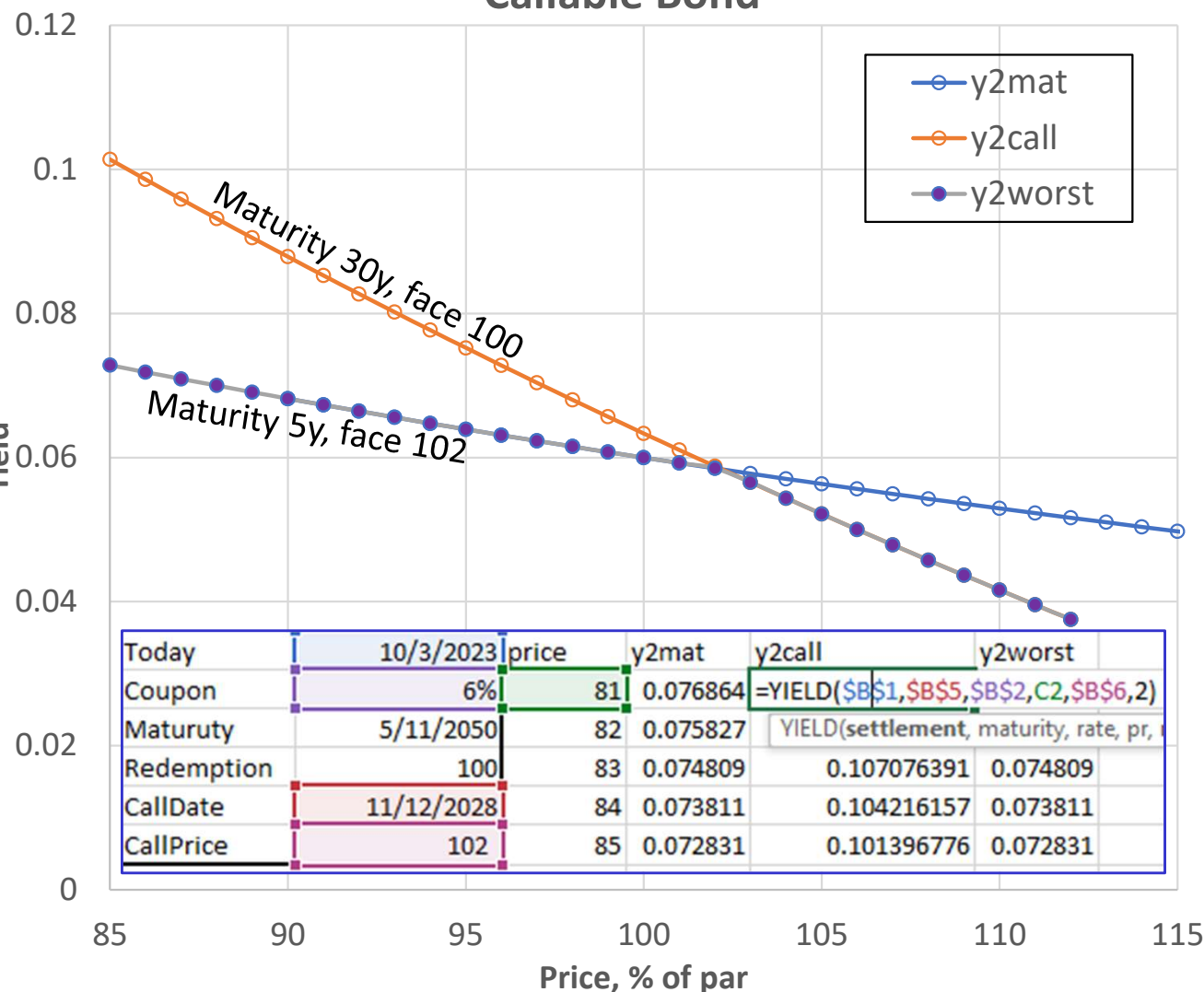
10 year bullet bond 5% coupon trading at par



- “Duration-neutral portfolio”
- “Hedge spread duration”

Callable bonds

Callable Bond



- Issuers exercise option when they can refinance cheaper, that is paying lower yield
- Investors need to be compensated for the loss of income; callable bonds are cheaper.
- Calls can be discrete, more than one, and with different prices
- Kink in the price-yield curve is not quote right → OAS
- Options-Adjusted Spread (or yield) is not observable but provides a framework for pricing optionality in bonds.
- Call date can be M-30, just an accounting trick allowing to call a bond at maturity.
- Make-Whole is a “mild” option that allows an issuer to call a bond but at a premium to compensate bond-holders.

Practical details

They call it “fixed income” for a reason

Rarely compute cash flows from scratch: QuantLib, YieldBook, Bloomberg, etc

<https://quantlib-python-docs.readthedocs.io/en/latest/dates.html>

- Calendar Types, Holidays, treatment of the end of months etc
 - Actual/Actual - Treasuries
 - 30/360 -- US corporate bonds and many US agency issues
- Settlement
 - T+1 Treasury
 - T+2 Corp, Muni
- Accruals
 - Coupon freq
 - Schedule
 - Clean price
 - For quoting
 - Dirty price
 - Settlement
 - Daily PNL
 - Coupon/360

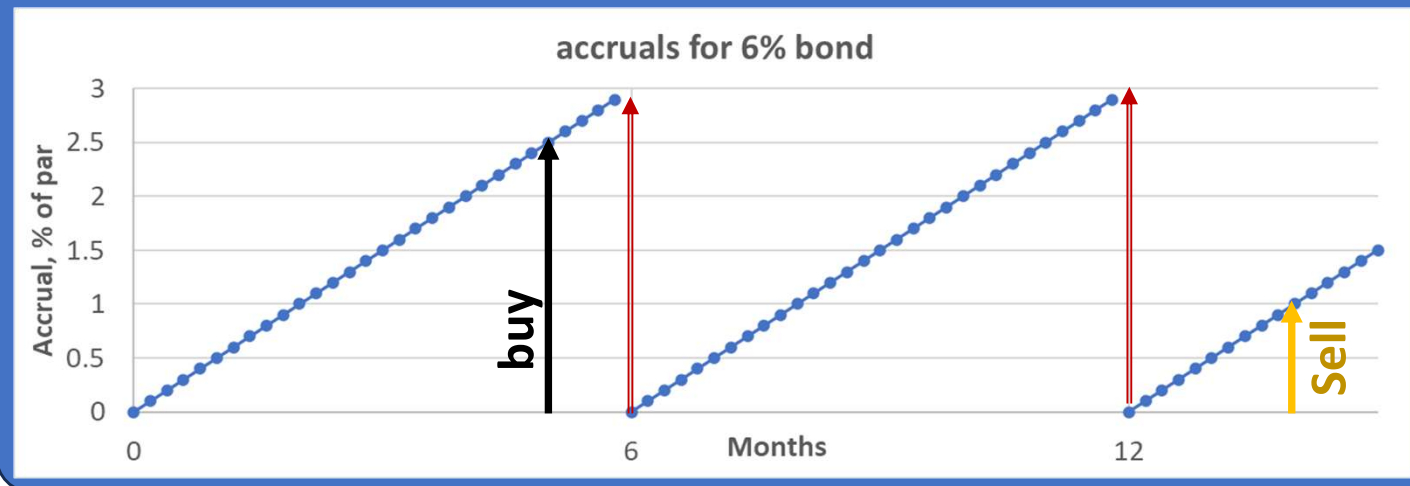
Clean price vs Dirty price

Buy 5 months after coupon @ 100 clean → 102.5 dirty

Get 2 coupons 6/2

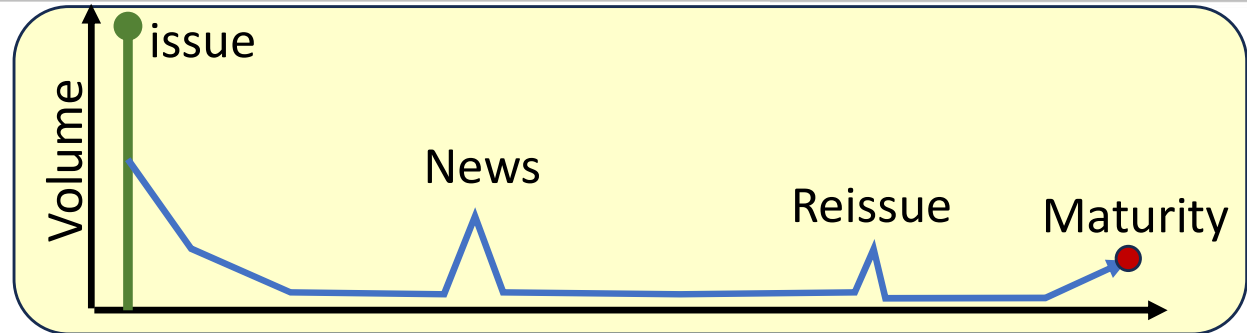
Sell 2 months after coupon @ 100 clean → 101.0 dirty

What if you buy 1 day before coupon date?

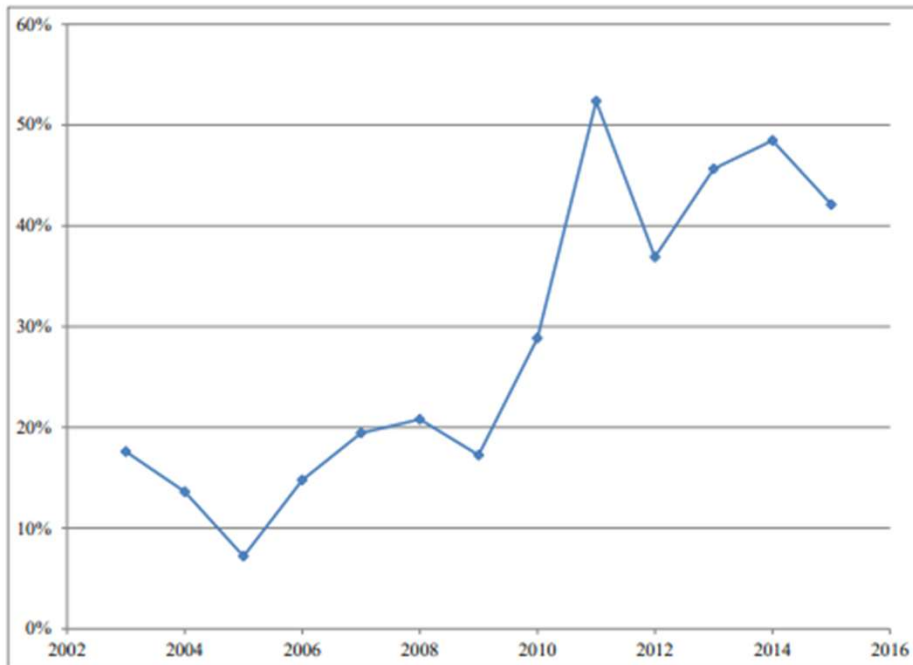


Bond Lifecycle

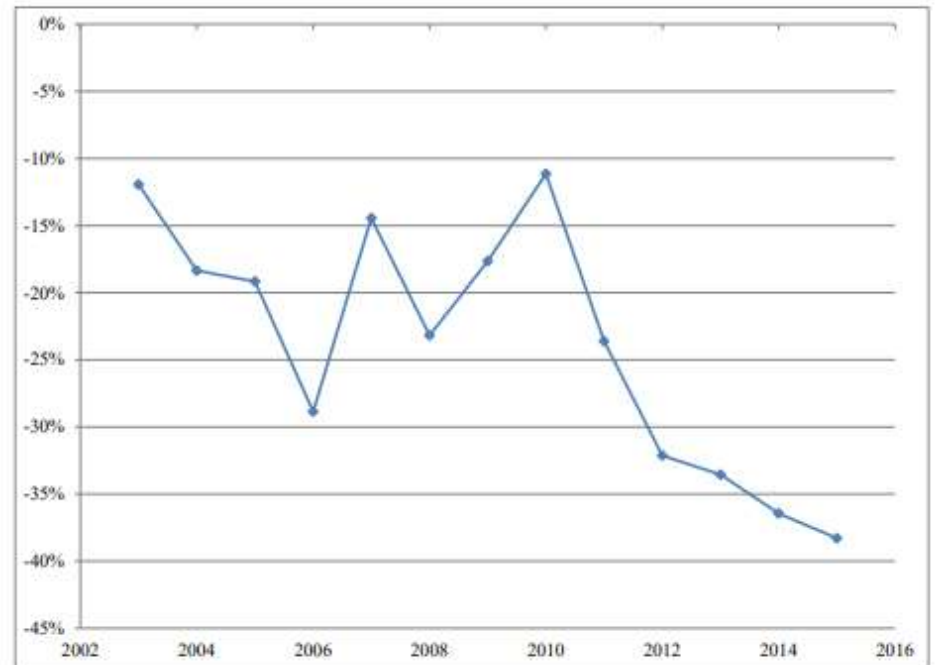
1. Issued
2. Actively traded
3. Parked
4. Taken out for rebalance
5. Parked
6. **Mature**



Percentage of Most Actively Traded Bonds Issued in the Last 90 Days



Trading Volume Reduction After 90 Days in the Secondary Market



"Analysis of Corporate Bond Liquidity" from Finra

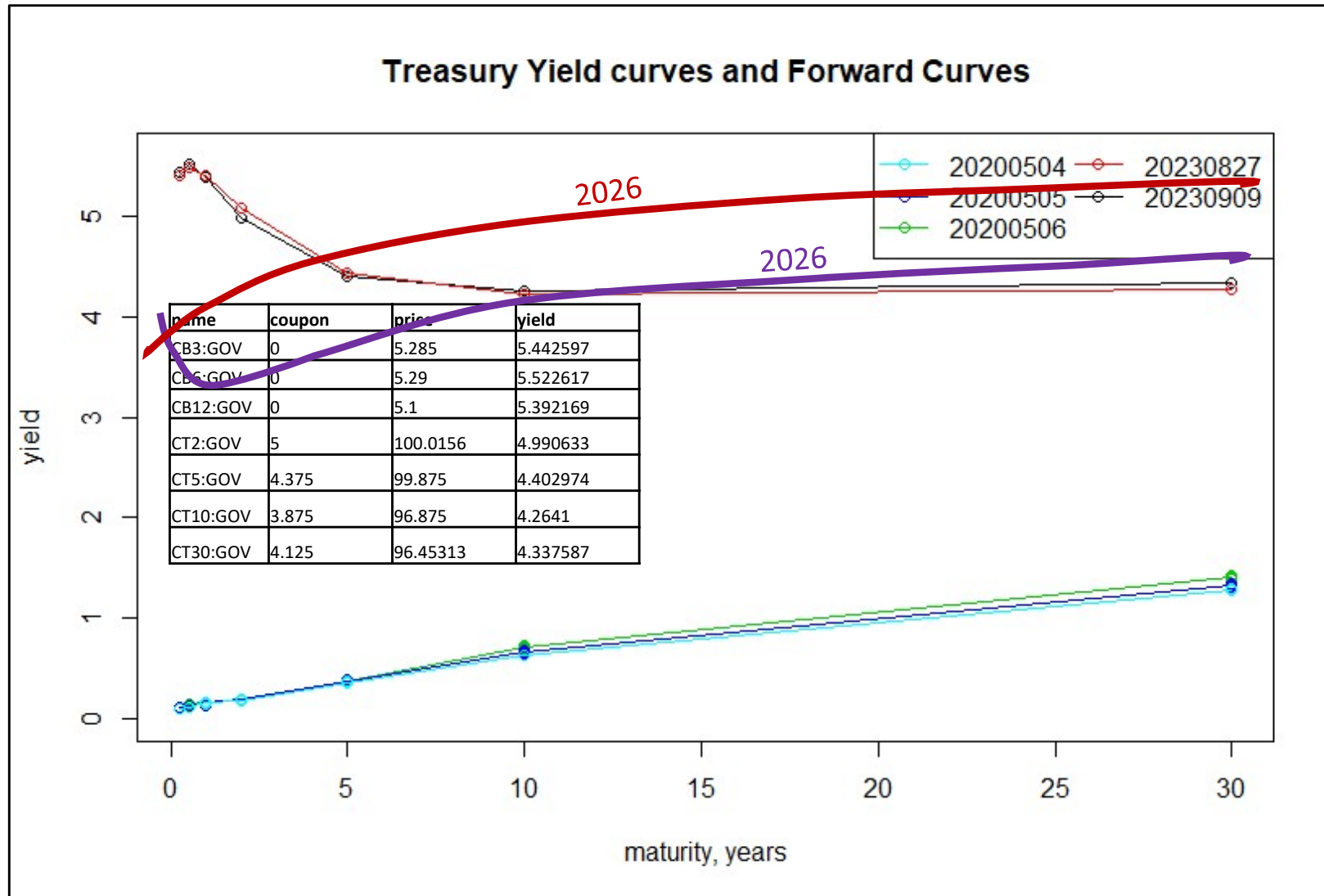
Bond Lifecycle – TSY Issuance



Treasury Yield Curves

Current bond yields

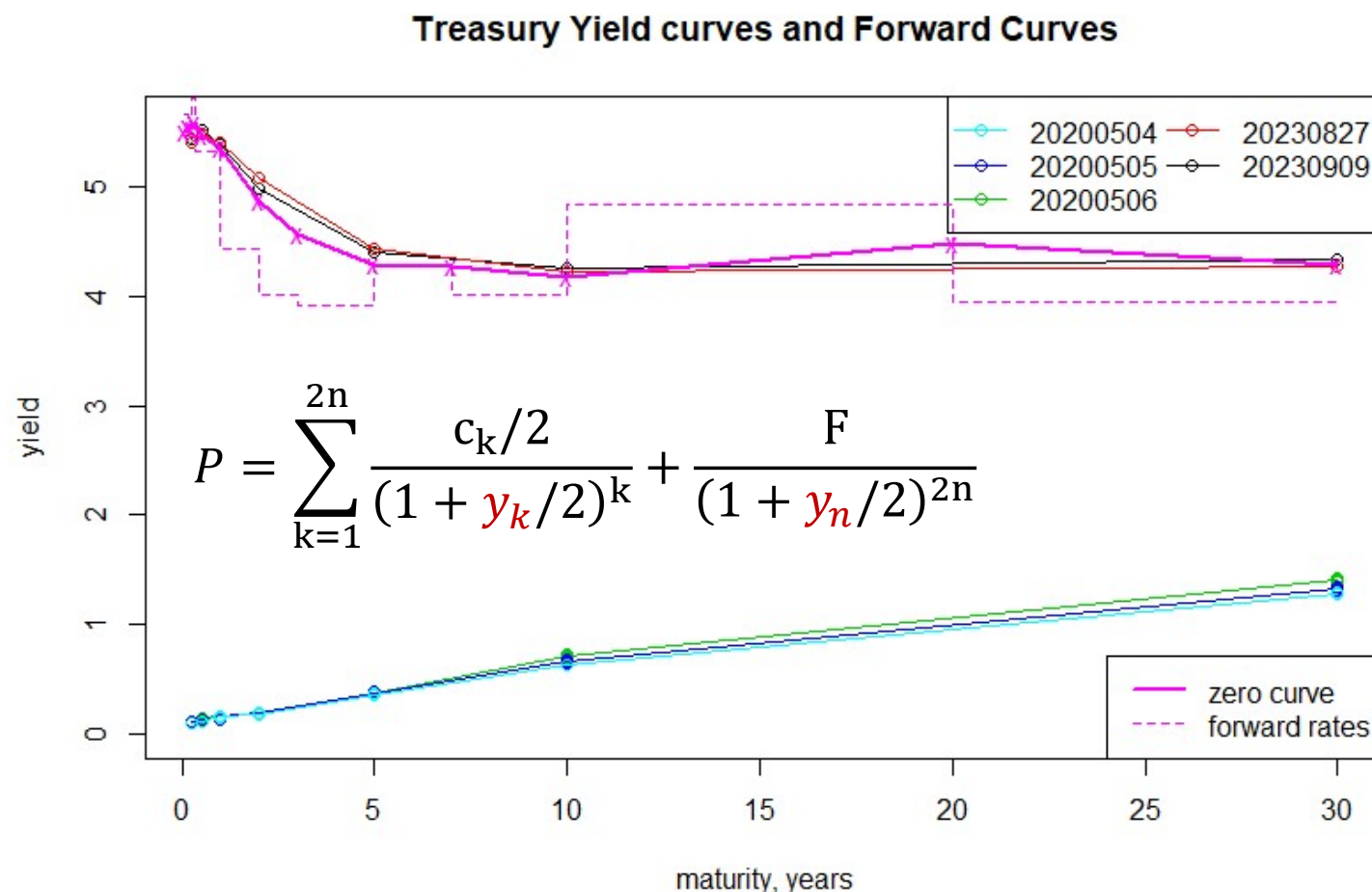
https://www.bloomberg.com/markets/api/comparison/data?securities=GB3%3AGOV,GB6%3AGOV,GB12%3AGOV,GT2%3AGOV,GT5%3AGOV,GT10%3AGOV,GT30%3AGOV&securityType=GOVERNMENT_BOND_US&locale=en



Collection of Treasury bonds with yields produces a zero-curve, e.g. by bootstrapping.

Zero yields

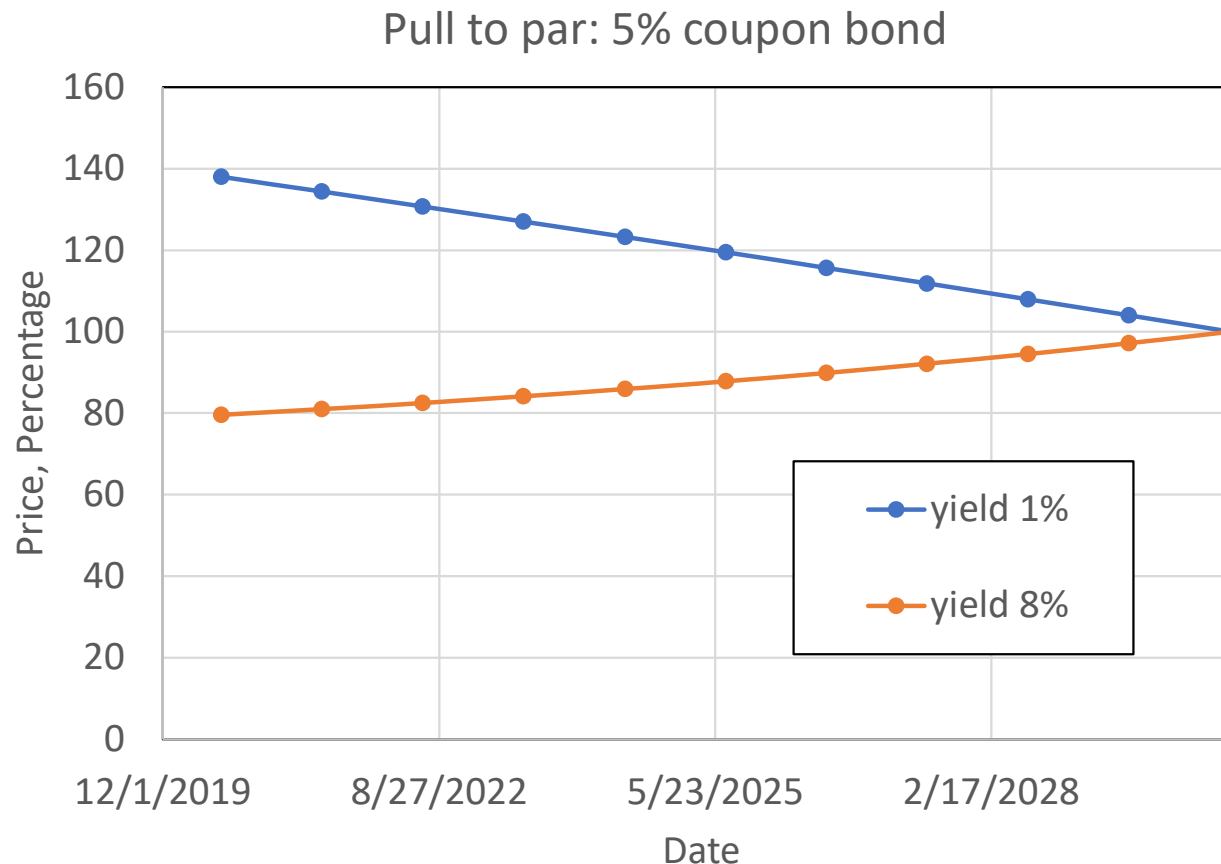
https://home.treasury.gov/resource-center/data-chart-center/interest-rates/TextView?type=daily_treasury_yield_curve&field_tdr_date_value_month=202308



Moving around a curve: carry

Carry: expected profit and loss (P&L) assuming yield of the bond remaining fixed.

- For a bond trading at par carry is the coupon,
- for a bond trading > 100 the bond price will decrease
- for a bond trading < 100 it will increase because of the pull-to-par-effect.
- No curve implied in carry calculation



Moving around a curve: Roll

Rolldown PL reflects yield change from maturity reduction, with fixed yield curve

- For regular upward-sloping curve the yield reduces, and the rolldown is positive
- For inverted curve it will be negative. Carry and rolldown are often taken together and represent the total PL arising from ageing of the bond while the curve remains fixed.
- $P(0) = \int_0^T D(t)c(t)dt + FD(T)$ price now, D stands for discount factor (not duration)
- $P(\Delta T) = \int_0^{T-\Delta T} D(t)c(t)dt + FD(T - \Delta T) \approx P(T) - cD(T)\Delta T - F \frac{dD}{dT} \Delta T$
- $\Delta P = -cD(T)\Delta T - F \frac{dD}{dT} \Delta T$

Credit relative value (RV) expresses the bond value relative to its peers.

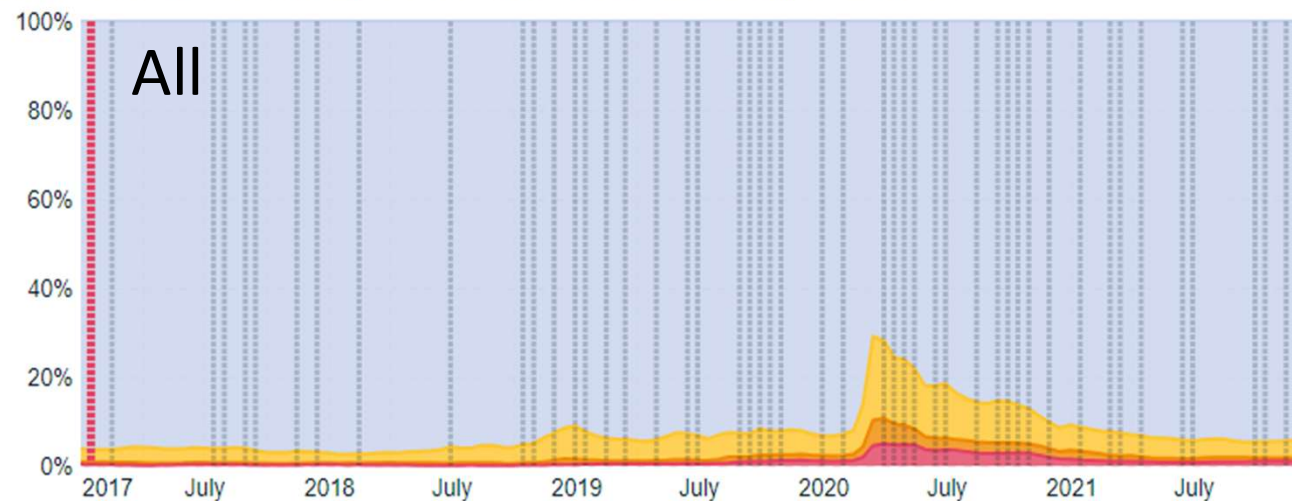
- Often referred to as 'rich/cheap to the curve' — implying a curve.

Rolldown and RV require the construction of an issuer's curve.

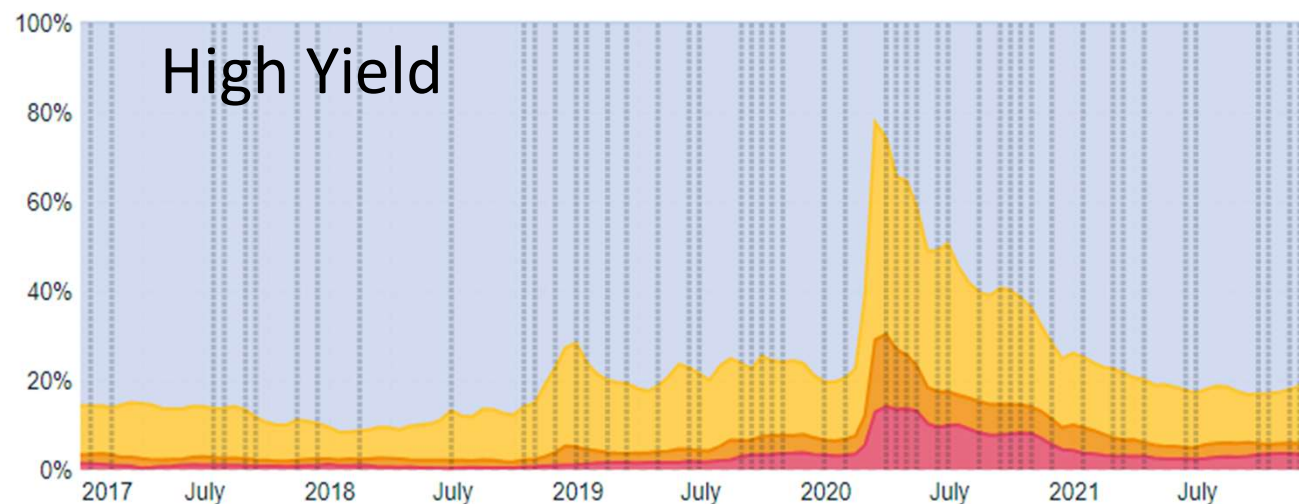
What if something goes wrong

Code	S&P	Moody's	Fitch
1	AAA	Aaa	AAA
2	AA+	Aa1	AA+
3	AA	Aa2	AA
4	AA-	Aa3	AA-
5	A+	A1	A+
6	A	A2	A
7	A-	A3	A-
8	BBB+	Baa1	BBB+
9	BBB	Baa2	BBB
10	BBB-	Baa3	BBB-
11	BB+	Ba1	BB+
12	BB	Ba2	BB
13	BB-	Ba3	BB-
14	B+	B1	B+
15	B	B2	B
16	B-	B3	B-
17	CCC+	Caa1	CCC+
18	CCC	Caa2	CCC
19	CCC-	Caa3	CCC-
20	CC	Ca1	CC
21	C	Ca2	C
22		C	DDD
23			DD
24	D		D
25	NR	NR	NR

Distribution of Issuers by 3Y Risk-Neutral Probability of Default



Distribution of Issuers by 3Y Risk-Neutral Probability of Default



<https://www.msci.com/research-and-insights/insights-gallery/probabilities-of-default-in-the-corporate-bond-market>

- Default-free DCF pricing

$$P(k = 1, n) = \sum_{k=1}^n \frac{C_k}{(1 + y_k/2)^k} + \frac{F}{(1 + y_n/2)^n}$$

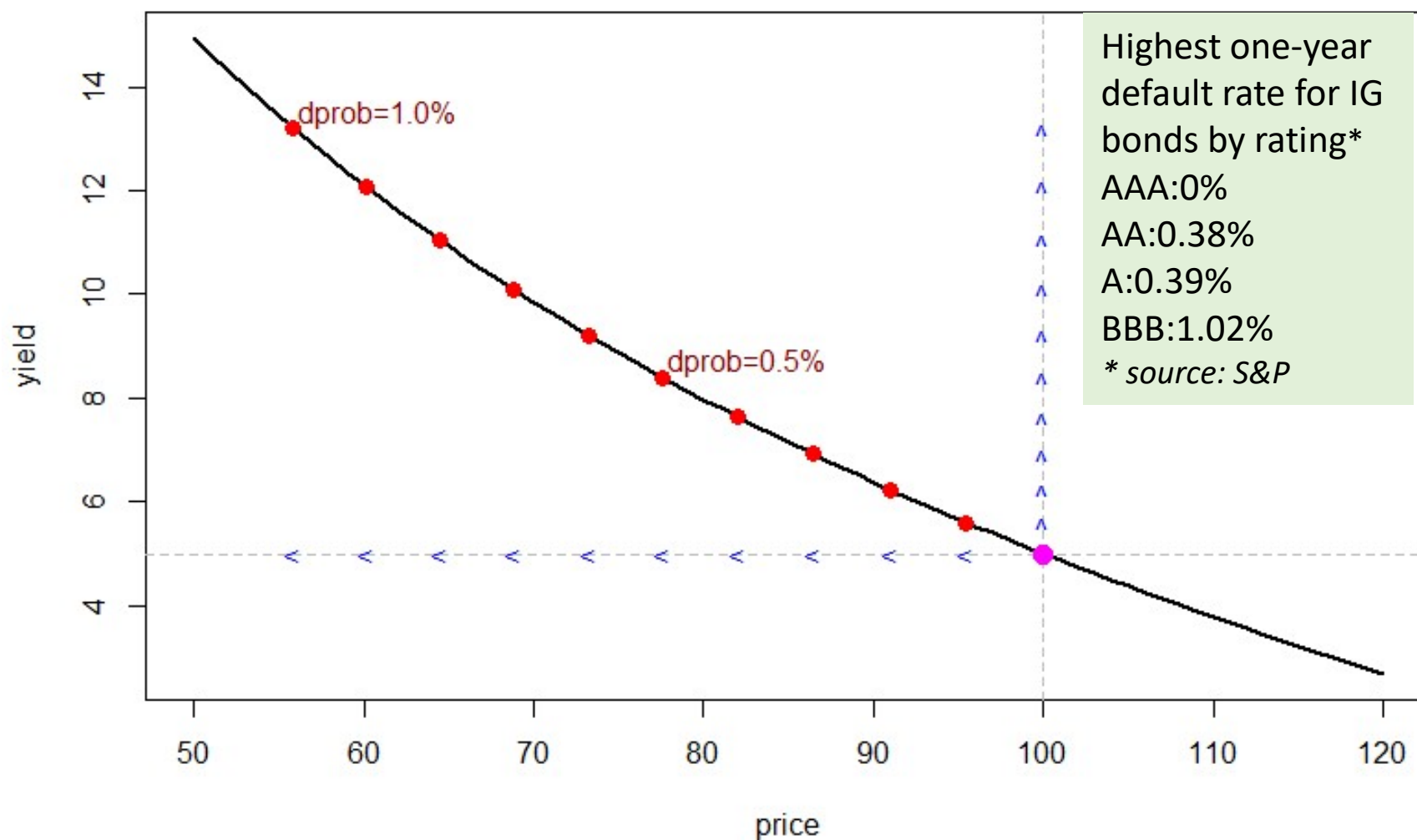
- Assume default rate d per period.
 - If defaulting at time k , is losing future cash flows.
 - Probability of default in period k but not sooner $d(1 - d)^{k-1}$
 - Assume zero recovery rate. Normally 40-50% recovery

$$P_D = P(n) - \sum_{k=1}^n d(1 - d)^{k-1} P(k, n)$$

For simplicity assume flat yield curve with $Y_{\text{zero}}(t)=5\%$

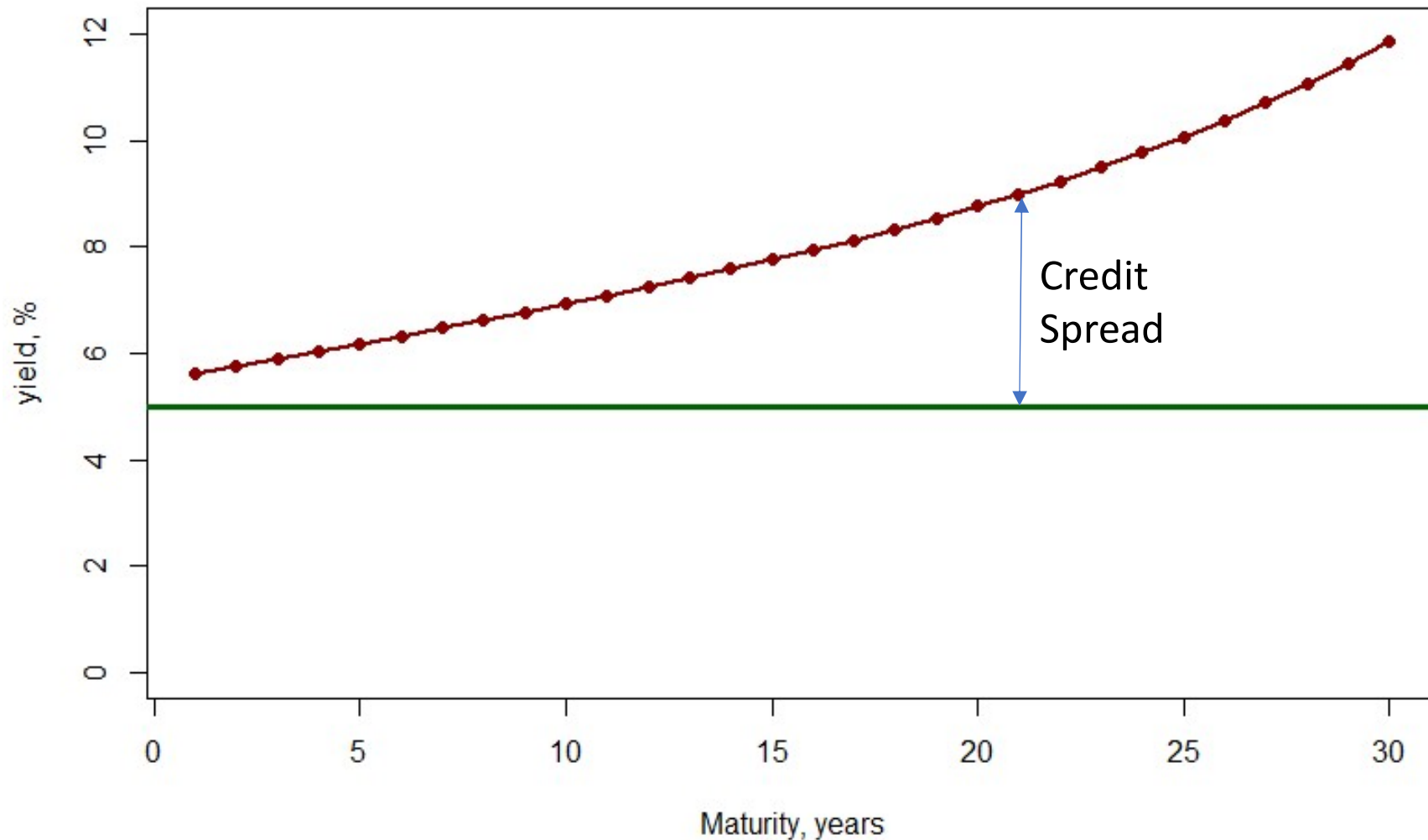
Pricing 10 year bullet bond with probability of default

bullet bond 5% coupon trading at par



Higher yield needed to compensate for expected default loss

Yield vs maturity for default prob=0.3% per year



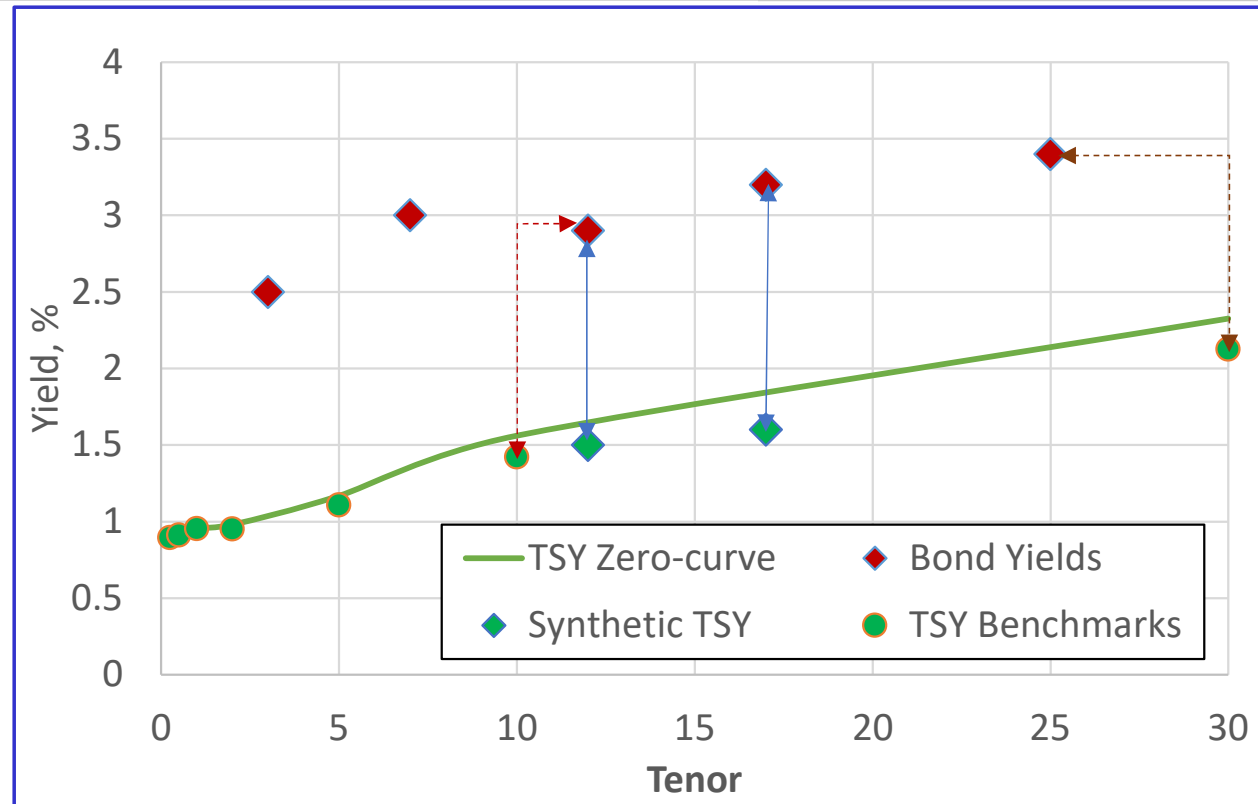
[A Practical Guide to the Valuation of Coupon-Bearing Fixed Income Securities](#), RA. Jarrow and DR. van Deventer (Kamakura)

Credit spreads

- Spread to treasury bid: $Y_{bond} - Y_{bm}$
 - Y_{bm} nearest on-the-run (recently issued), aka OTR, treasury
 - No curves involved – simple
 - Widely used for quoting
 - Maturity mismatch
 - Jumps when new treasuries are issued and benchmark changes

<18	[18, 48]	[48, 72]	[72, 96]	[96,180]	>=180
CT2	CT3	CT5	CT7	CT10	CT30B

- G-spread: spread to the interpolated treasury curve (I25)
 - Yield – T-YieldCurve
 - Independent of individual TSY behavior.
- I-spread - to the CDS curve (\$490 USD SOFR)
- Asset Swap Curve spread (ASW)
 - <https://cms.prod.bloomberg.com/team/pages/viewpage.action?pageId=466542223>



- “H”-spread: spread to synthetic (hypothetical) treasury
 - Yield - DCF(zero t-curve, cashflows)
 - Use of the yield curve
 - Independent of individual TSY behavior.

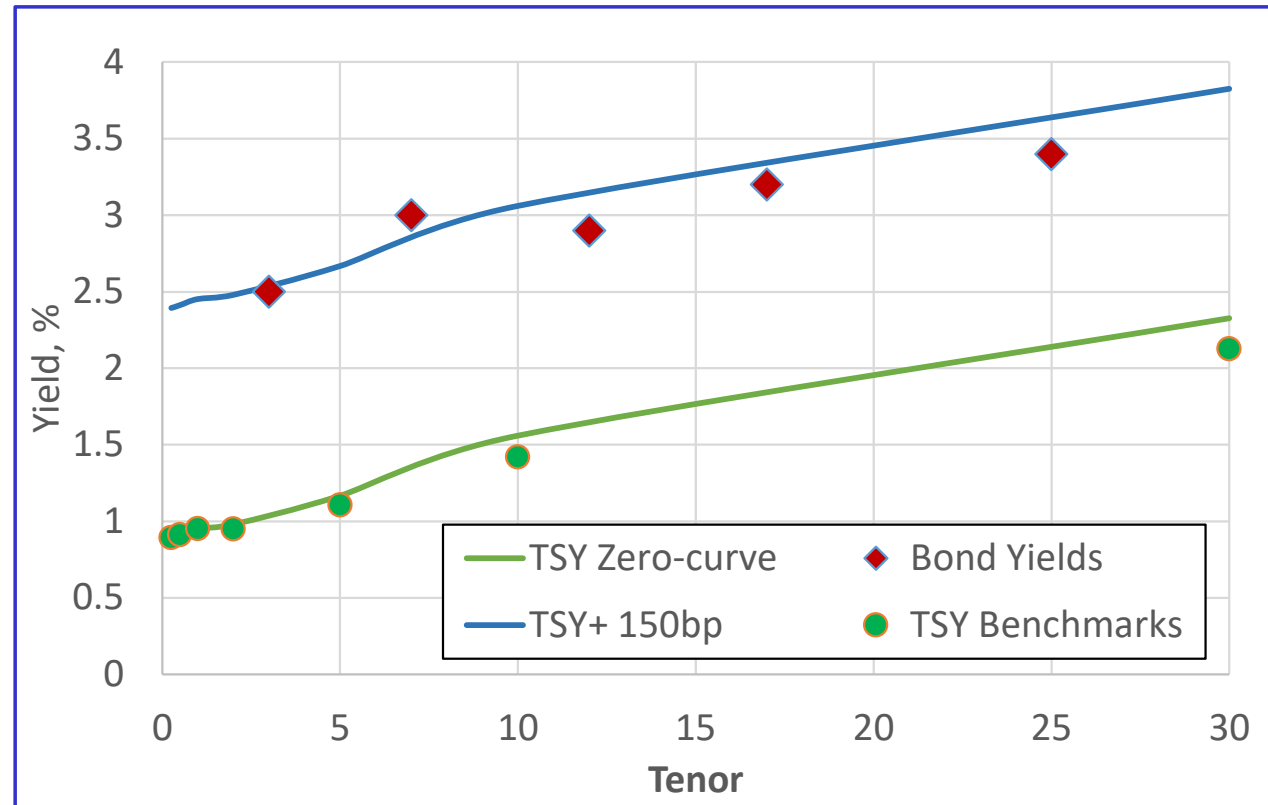
Credit spreads

- Z-spread or zero-curve spread

- By how much I need to shift zero-coupon treasury curve y_k to get current bond price/yield
- Useful for comparing bonds and simple models of bond spreads (with a constant)

- $$P = \sum_{k=1}^n \frac{C_k}{(1+(\mathbf{z}+y_k)/2)^k} + \frac{F}{(1+(\mathbf{z}+y_n)/2)^n}$$

- Solve for $\mathbf{z}$, given P , and y_k
- Z-spread is bond-specific



[The credit spread curve I: Fundamental concepts, fitting, par-adjusted spread, and expected return](#) by R.J. Martin (2022)

Security Master

- Bloomberg back-office
- ThomsonReuters aka Refinitiv (LSEG)
- Venues (subsets of instruments and fields)
- Exchange-Traded Funds

FIGI	Name	Ticker ↑	Exchange Code	Security Type
Filter	Filter	Filter	Filter	Filter
BBG001B59334	ANHEUSER-BUSCH INBEV WOR	ABIBB 3.65 01/15/16 MPLE	NOT LISTED	PRIV PLACEMENT
BBG00BW1Q8V7	ANHEUSER-BUSCH INBEV FIN	ABIBB 3.65 02/01/26	TRACE	GLOBAL
BBG00NWNWY25	ANHEUSER-BUSCH CO/INBEV	ABIBB 3.65 02/01/26 *	NEW YORK	GLOBAL
BBG00MJFVPK5	ANHEUSER-BUSCH CO/INBEV	ABIBB 3.65 02/01/26 144A	TRACE	PRIV PLACEMENT
BBG00MJFVV59	ANHEUSER-BUSCH CO/INBEV	ABIBB 3.65 02/01/26 REGS	TRACE	EURO-DOLLAR

<https://www.openfigi.com/search#!?page=1&simpleSearchString=anheuser&marketSector=Corp&facets=>

Bond Symbology

- CUSIP (US+Canada) Equities, Credit, Treasuries
 - Entity(numeric 6) + instrument(charnum 2) + checksum(1) : 035242AP1
 - Tends to run out, messed up by M&A
 - Can be all numbers, messes up parsers
 - “9128” US Treasury
- ISIN (international)
 - Country(2) + InternalID (9) + checksum(1): US035242AP13
 - Incorporates CUSIP for US and CA
 - <https://www.boerse-frankfurt.de/bonds/search>
- CINS
 - Cusip-like format for non-US instruments
- SEDOL
- FIGI
 - BBG00BW1Q8V7 free in openFIGI API
 - <https://www.openfigi.com/search#!?page=1&simpleSearchString=US035242AP13>

Do secmaster fields matter?

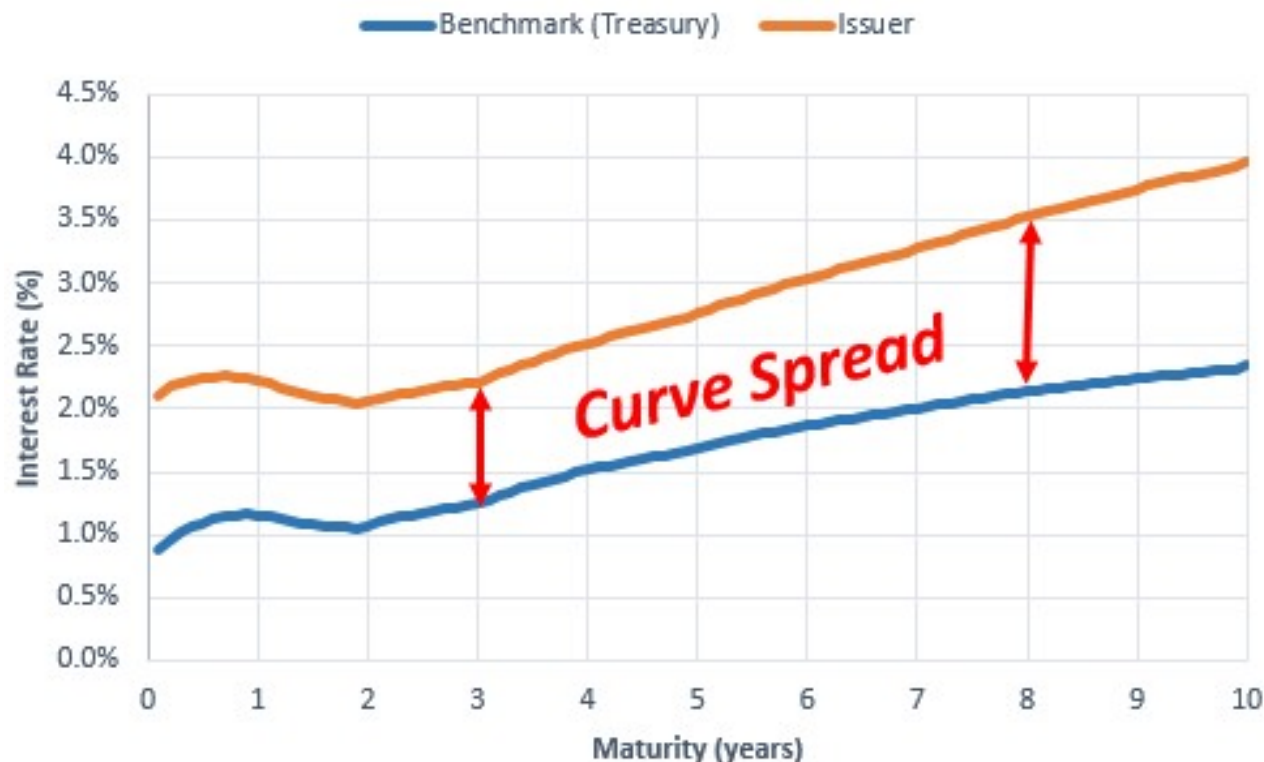
- Issuer – Ford vs Ford Fin Svs
- Callable/puttable/make-whole bond?
 - Call schedule `[(date1,price1), (date2,price2),...]`
 - Call at Maturity-30d
 - Call notice period
 - Called (yes/no)
- RegS / 144A – issues for institutional investors
- Coupon
 - Step-up, first coupon payment, long/short first|last, floating rate
- Sinkable
- Insured, asset-backed
- Issue date, issueSize, amountOutstanding, float
- Bond rating; place in capital structure (Senior/Junior)
- minSize, minIncrement → Can be overloaded by venue or traders

What affects bonds prices?

- 5-6-7-8 Factor models
 - <https://faculty.georgetown.edu/qw50/RiskFactor.pdf>
 - downside risk, credit risk, and liquidity risk positively predict the cross-sectional variation in future bond returns.
- Trade Price Clustering
 - <https://www.emerald.com/insight/content/doi/10.1108/CFRI-02-2022-0013/full/pdf?title=trade-price-clustering-in-the-corporate-bond-market>

How do we know default prob? ...and do we need to know it?

- Ratings (Moody's, S&P, Fitch)
 - Derived from capital structure
<https://www.moodys.com/sites/products/defaultresearch/2006200000425249.pdf>
- Market Pricing
 - How much cheaper is a bond than a risk-free bond in the market?



Systematic trading? Let's try to build a corporate bond trading system.

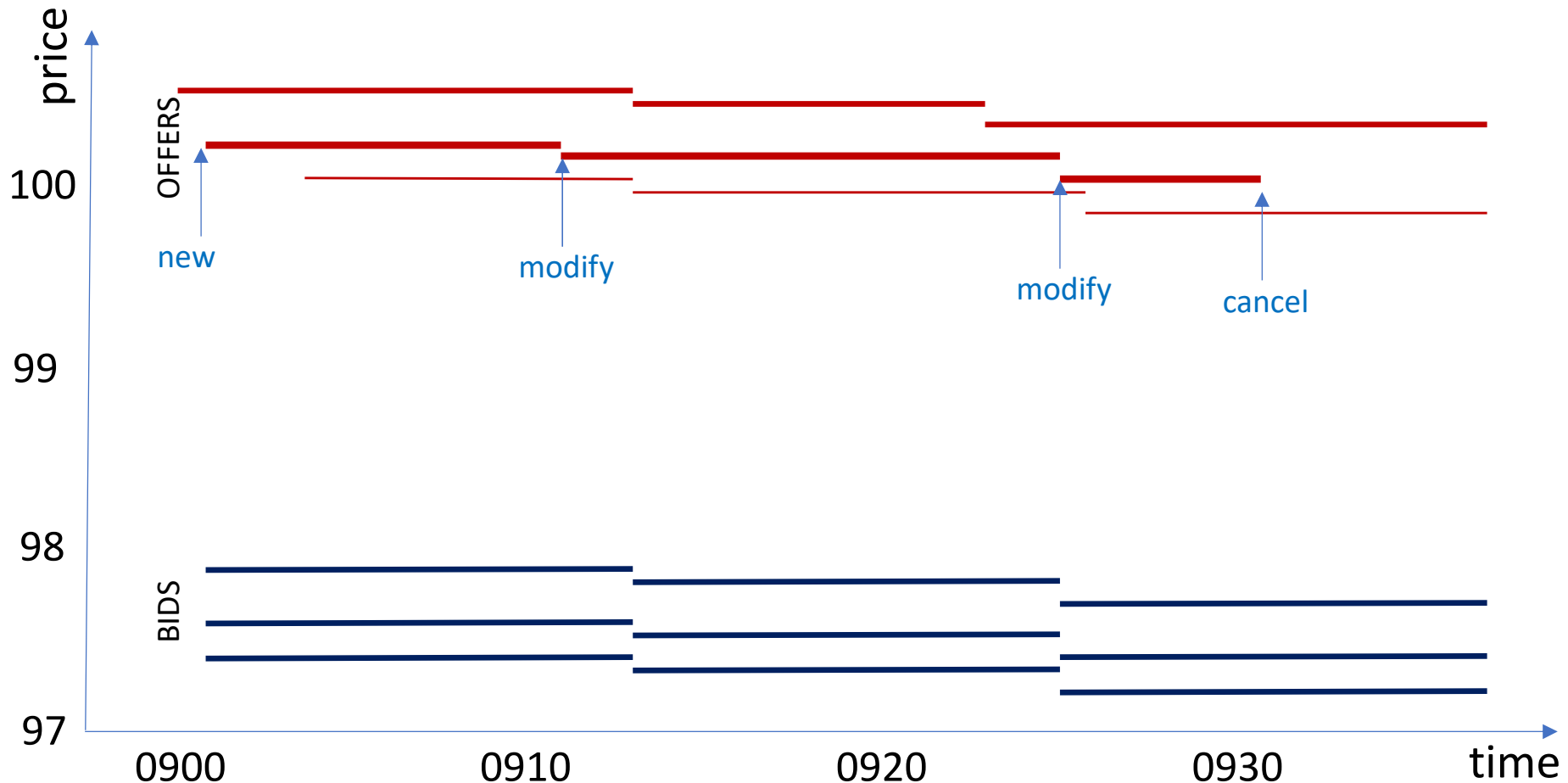
- Green-field vs legacy?
- List of instruments
- Market Data – can be voluminous
- Model
 - Signals. What time scale?
 - Risk model / inventory management
 - Quoting/trading
- Strictly speaking Systematic != Automated
- Connectivity to the venues or clients
- Reporting (sell-side)
- Allocation, constraints (buy-side)

Market Data

- Distributed over FIX (Financial Information eXchange)
 - tag=value text based
 - 10k msg/second
 - <https://fixparser.targetcompid.com/> for quick debugs
- Order, IOI,
 - datetime, symbol, side, **action**, price, **pType**, msgId, [refMsgId]
 - **Action** = [New(Add), Change(Modify), Delete(Cancel)]
 - **pType** = [Percentage, Yield(yieldType), Spread (benchmark)]
- Quote
 - datetime, symbol, bid, bidSize, ask, askSize, **pType**, quotId
 - priceType = [Percentage, Yield(yieldType), Spread (benchmark)]
- Permissioning – not everyone sees everyone quotes

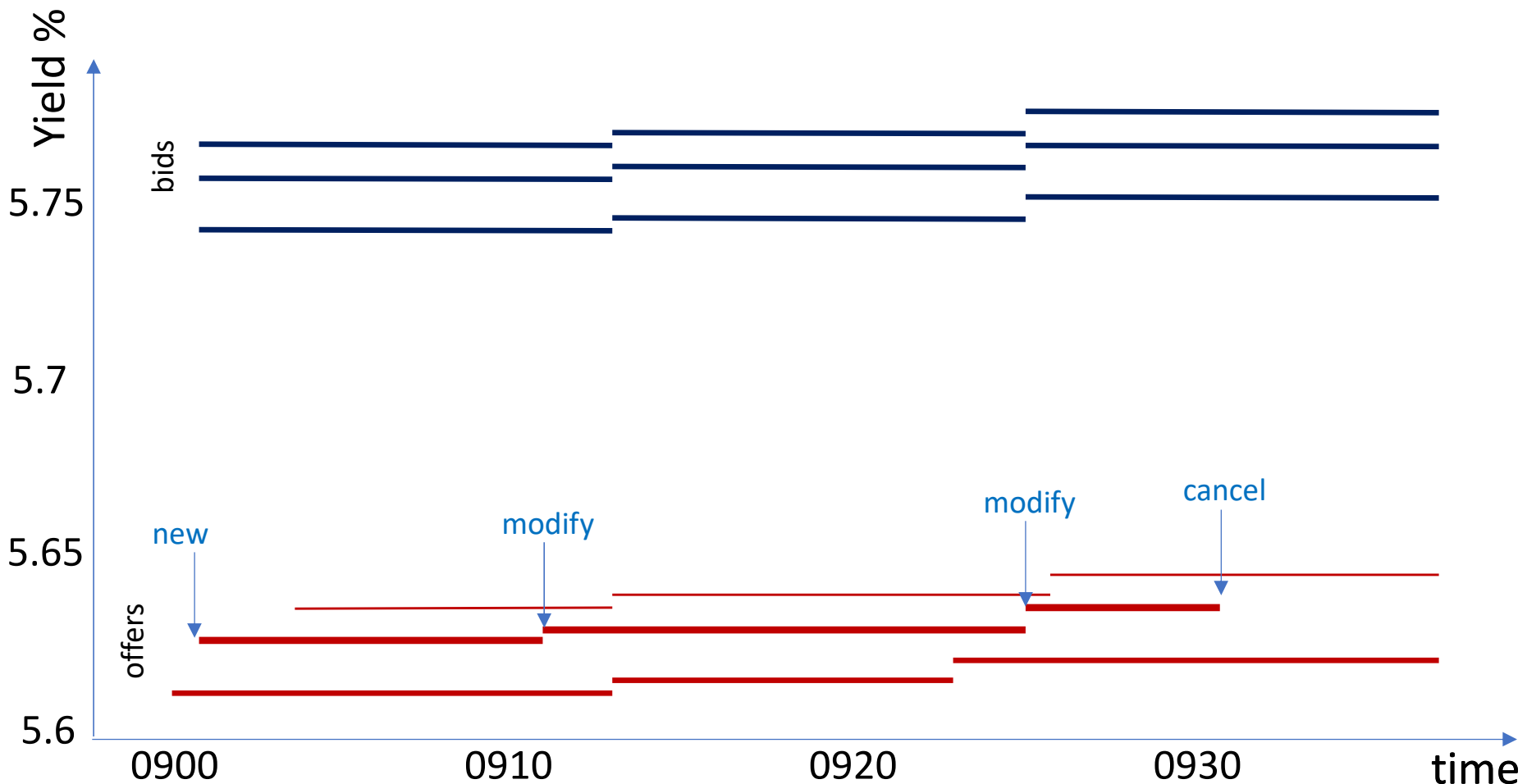
Market Data: quote and orders

09:03:13 symbol=00206RBH4 side=S price=100.300 size=75 action=ADD id=12
 09:13:21 symbol=00206RBH4 side=S price=100.255 size=75 action=MODIFY id=13 ref=12
 09:25:21 symbol=00206RBH4 side=S price=100.155 size=75 action=MODIFY id=14 ref=12
 09:31:44 symbol=00206RBH4 side=S price=100.250 size=75 action=CANCEL id=15 ref=12



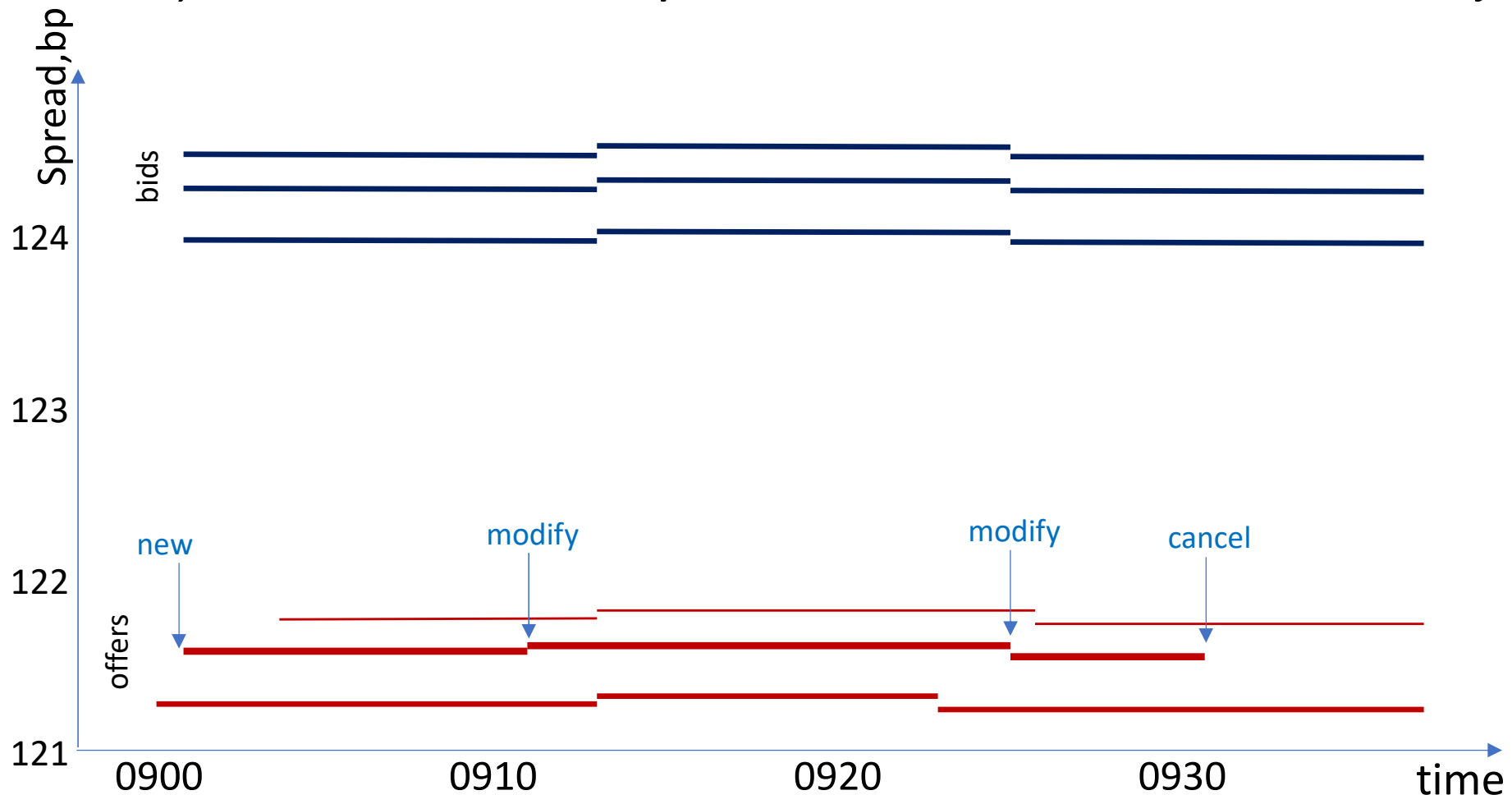
Market Data: order book

09:03:13 symbol=00206RBH4 side=S **yield=5.627** size=75 action=ADD id=12
 09:13:21 symbol=00206RBH4 side=S **yield=5.627** size=75 action=MODIFY id=13 ref=12
 09:25:21 symbol=00206RBH4 side=S **yield=5.631** size=75 action=MODIFY id=14 ref=12
 09:31:44 symbol=00206RBH4 side=S **yield=5.631** size=75 action=CANCEL id=15 ref=12



Market Data: order book

09:03:13 symbol=00206RBH4 side=S **spread=121.5** size=75 action=ADD id=12
 09:13:21 symbol=00206RBH4 side=S **spread =121.6** size=75 action=MODIFY id=13 ref=12
 09:25:21 symbol=00206RBH4 side=S **spread =121.5** size=75 action=MODIFY id=14 ref=12
 09:31:44 symbol=00206RBH4 side=S **spread =121.5** size=75 action=CANCEL id=15 ref=12



TRACE

Trade Reporting and Compliance Engine

<https://www.finra.org/filing-reporting/trace>

<http://www.finra.org/sites/default/files/BTDS-specs-v4.4.pdf>

- Introduced 2003
- Documented evidence of improved transparency (e.g. lower bid-offer spread)*
- Expanded to structured products and 2012, treasuries 2023
- Portfolio Trading indicator for corporate bonds added May 2024

Each dealer or market venue trade must report its trades to FINRA

RPS – Reporting Party Side [**B**uy|**S**ell]

RPT – Reporting Party Type [**D**ealer]

CPT – Counterparty Type [**C**lient|**A**ffiliate] buy side and [**D**ealer|**A**TS] sell side

ATS – Automated Trading System [YES|*blank*]

Clean-up

- For each cusip order transactions by size, then execution time.
- For matching size and ts, “D”+”S”=”S”; D+D+S=S; S+B=D etc.
- Mark “compressed” entries.

Derived data for modeling and universe filtering (examples)

- Compute daily volume on cleaned data
- Compute daily ADV and ADV-250 (retail trades) files.
- Trade imbalance (buy minus sell)

[Capital Commitment and Illiquidity in Corporate Bonds](#) Hendrik Bessembinder et al, 2016

TRACE + Market Data.

RPS – Reporting Party Side [**B**uy|**S**ell]

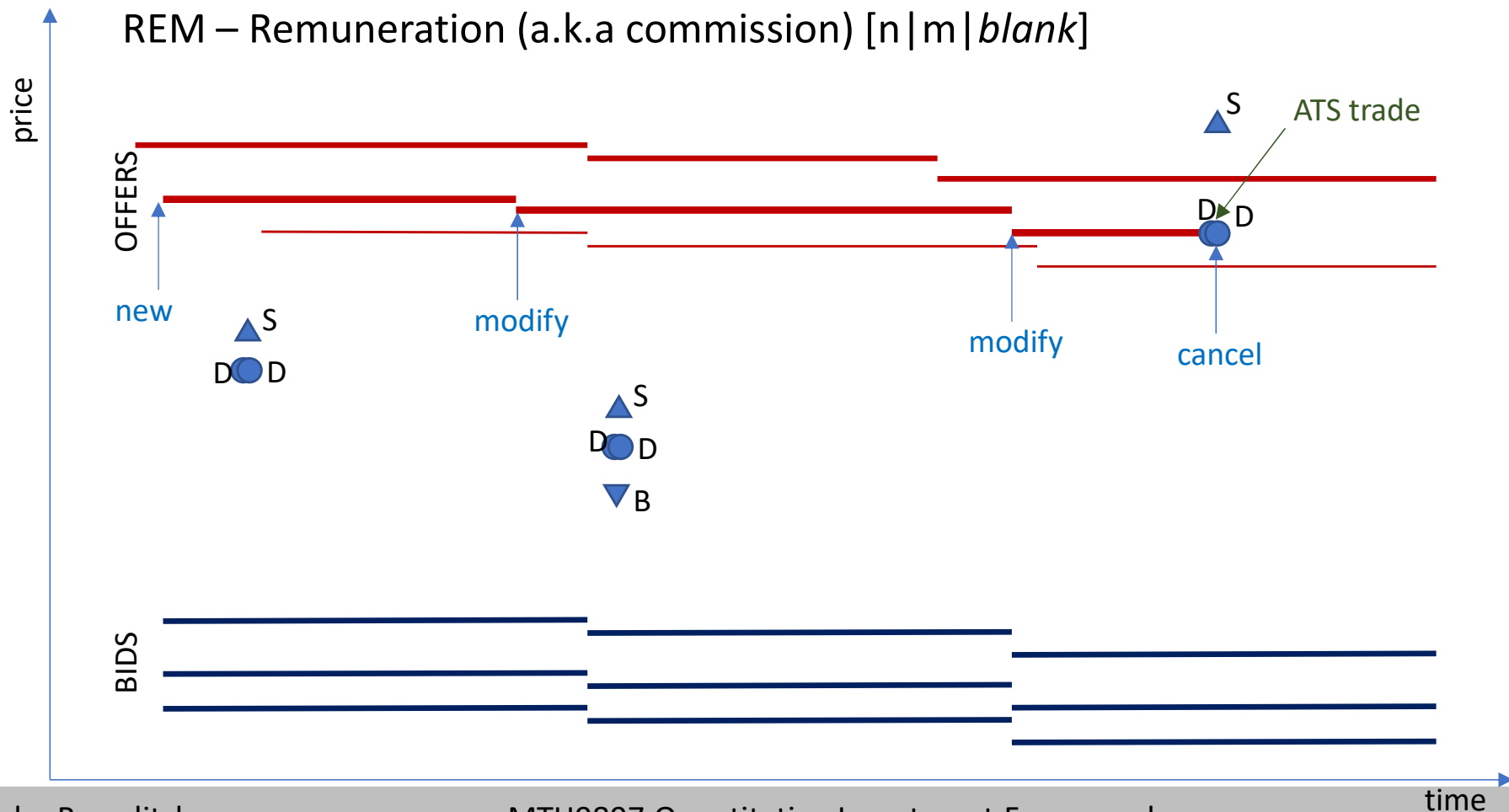
RPT – Reporting Party Type [**D**ealer]

CPT – Counterparty Type [**C**lient|**A**ffiliate|**D**ealer|**ATS**]

for practical purposes C==A and D==T

ATS – Automated Trading System [YES|*blank*]

REM – Remuneration (a.k.a commission) [*n*|*m*|*blank*]



TRACE matching example

RPS – Reporting Party Side [**B**uy|**S**ell]

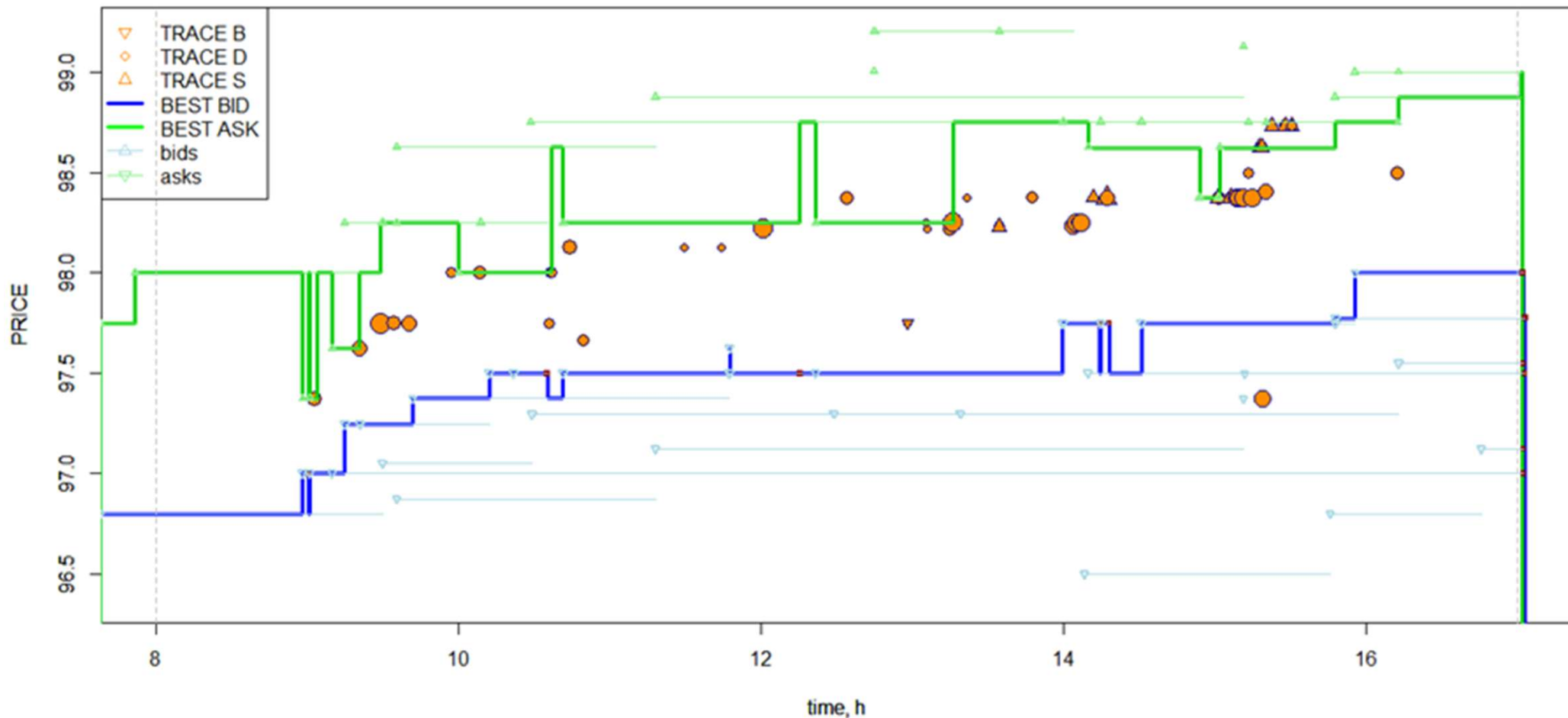
RPT – Reporting Party Type [**D**ealer]

CPT – Counterparty Type [**C**lient|**A**ffiliate|**D**ealer|**ATS**]

ATS – Automated Trading System [YES|*blank*]

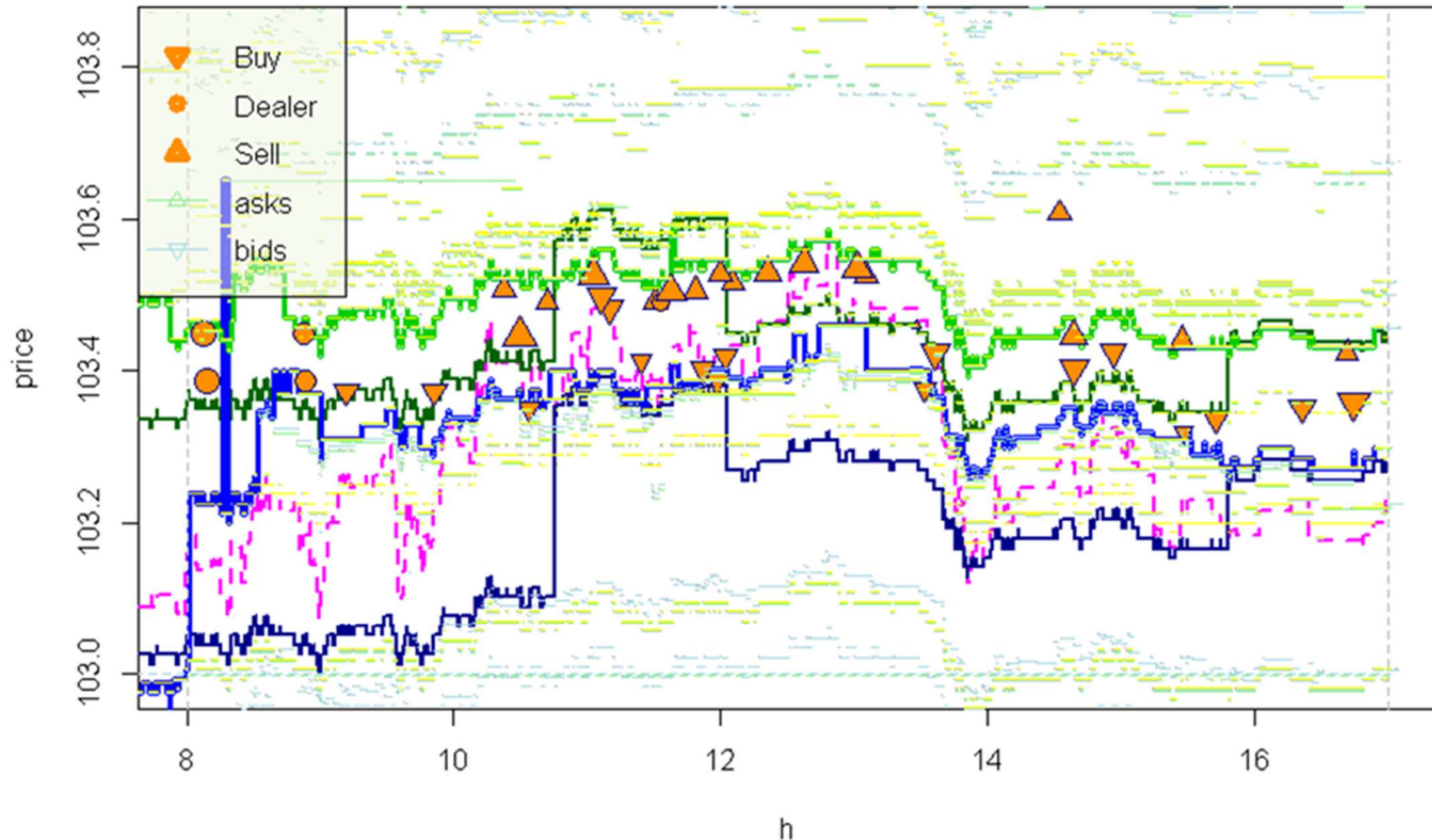
Date	Time	Size(M)	Price	Yield	RPS	RPT	CPT	Sprd	Trd Date	Trd Time	Rem	ATS	IGNORE	SIDE
11-Aug	16:14:21	3	99.95	N.A.	S	D	D	NA	11-Aug	16:14:12			NO	B
11-Aug	16:14:21	3	99.95	N.A.	B	D	C	NA	11-Aug	16:14:12	n		YES	
11-Aug	15:45:16	35	100.279	N.A.	S	D	C	NA	11-Aug	15:45:12	m			
11-Aug	14:29:41	5	100.205	N.A.	S	D	D	NA	11-Aug	14:29:41			NO	B
11-Aug	14:29:41	5	100.17	N.A.	B	D	C	NA	11-Aug	14:29:40	m		YES	
11-Aug	13:35:29	76	100.161	N.A.	B	D	C	NA	11-Aug	13:35:25	m			
11-Aug	11:24:02	55	100.117	N.A.	B	D	C	NA	11-Aug	11:23:42	n		YES	
11-Aug	11:24:02	55	100.117	N.A.	S	D	D	NA	11-Aug	11:23:42			NO	B
10-Aug	17:12:06	200	100.3	N.A.	S	D	C	NA	10-Aug	17:12:05	m			
10-Aug	14:29:05	500	100.228	N.A.	S	D	D	NA	10-Aug	14:27:00			???	???
10-Aug	14:22:09	13	100.26	N.A.	S	D	D	NA	10-Aug	14:21:31				
10-Aug	14:21:47	500	100.208	N.A.	S	D	D	NA	10-Aug	14:21:45			???	???
10-Aug	13:50:42	2	100.127	N.A.	B	D	C	NA	10-Aug	13:46:16	m			
10-Aug	10:31:34	5	100.309	N.A.	S	D	D	NA	10-Aug	10:31:01		Y	NO	S
10-Aug	10:31:05	15	100.309	N.A.	S	D	D	NA	10-Aug	10:30:43		Y	NO	S
10-Aug	10:31:01	5	100.309	N.A.	S	D	C	NA	10-Aug	10:31:00	n		YES	
10-Aug	10:30:42	15	100.309	N.A.	S	D	C	NA	10-Aug	10:30:40	n		YES	
10-Aug	10:14:14	530	100.322	N.A.	S	D	C	NA	10-Aug	10:13:12	m			
9-Aug	12:01:32	4	100.188	N.A.	S	D	C	NA	9-Aug	12:01:28	m			
9-Aug	11:56:20	10	100.02	N.A.	B	D	C	NA	9-Aug	11:56:19	n		YES	
9-Aug	11:56:20	10	100.02	N.A.	S	D	D	NA	9-Aug	11:56:19		Y	NO	B
9-Aug	11:52:55	120	100.188	N.A.	S	D	A	NA	9-Aug	11:52:55	n		NO	B
9-Aug	11:52:55	120	100.162	N.A.	B	D	C	NA	9-Aug	11:52:54	m		YES	
9-Aug	11:45:14	50	100.043	N.A.	S	D	D	NA	9-Aug	11:45:07		Y	NO	B
9-Aug	11:45:14	50	99.963	N.A.	B	D	C	NA	9-Aug	11:45:07	m		YES	

BBG005Q4TXF1



A day in life of an Anheuser-Busch bond (~2018)

ABIBB 3.65 02/26 (035242AP1)



Notes on market data

- Bond with 10 trades in a day is “liquid”
- Most bonds don’t trade on a given day
- Quotes can be subject/indicative or firm
- Firm (e.g. Bloomberg ‘magenta’ quotes) 90% firm
- Prices can be stale
- Spread quotes can be coming with different non-default benchmarks

<18	[18, 48)	[48, 72)	[72, 96)	[96,180)	>=180
CT2	CT3	CT5	CT7	CT10	CT30B

SYSTEMATIC TRADING MTH9897

Lecture 2: Fixed Income

Quantitative Investment Framework



Misha Boroditsky, PhD

Bloomberg LLC

mboroditsky@hotmail.com

Teaching Assistant:

Alan Coman alancoman@gmail.com

Trading mode (in order of complexity)

ATS Liquidity Taking (IBP, TMC, TWD, MTS, ALLQ)

Hit bid/lift offer →

← Get trade or DNT

RFQ Taking (MKTX, IBP, TMC, TWD, ALLQ, BOLT)

Send RFQ →

← Get Quote

Accept or DNT (*auto-accept) →

← Get trade

Auction (TruMid, LiquidNet, GFI)

Send Interest →

← Get matching interest

← Get trade

RFQ Providing anonymous (MKTX OT, IBP, TMC)

← Get RFQ

Send Quote →

← Client accepts or DNT (* auto-accept)

Get trade ←

Streaming Quotes Anonymous (MTS, IBP, TMC)

Send Quote →

Send Quote →

Send Quote →

← Get trade

Technology

Connectivity (FIX)

Logon/Logoff/Reconnect

Normalization to message schema {Order, RFQ, Quote, Trade}.

Kafka producer/Consumer

Trading engine -- real time event processor

Real time risk management

e.g. outstanding orders

trading controls

Load examples

TRACE (bond trades) < 100K / day

MSRB (muni trades) < 50K / day

FI RFQ ~ 100K / day

Runs/Axes ~ 200K / day

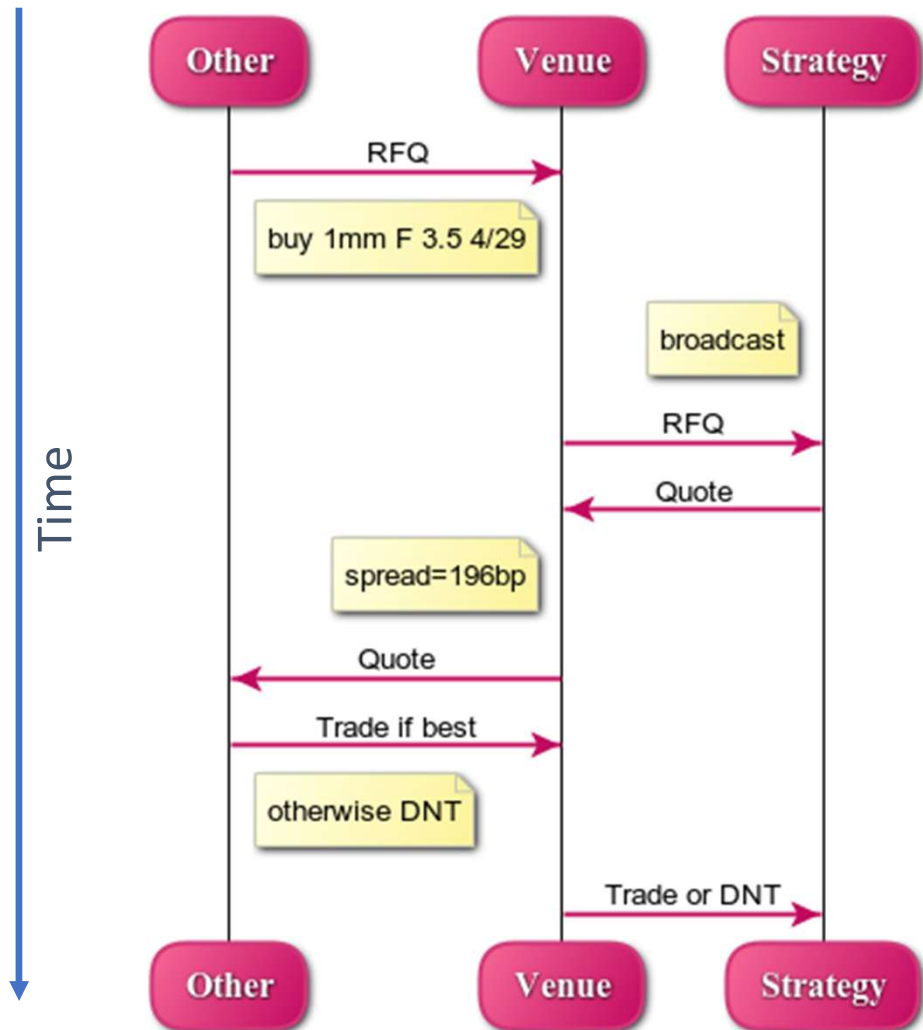
Evaluated CP+ ~ 1000/second

MarketData ~ 4000/second/venue

Processed market data ~ 300/sec

Request For Quote

Request For Quote (RFQ)



www.websequencediagrams.com

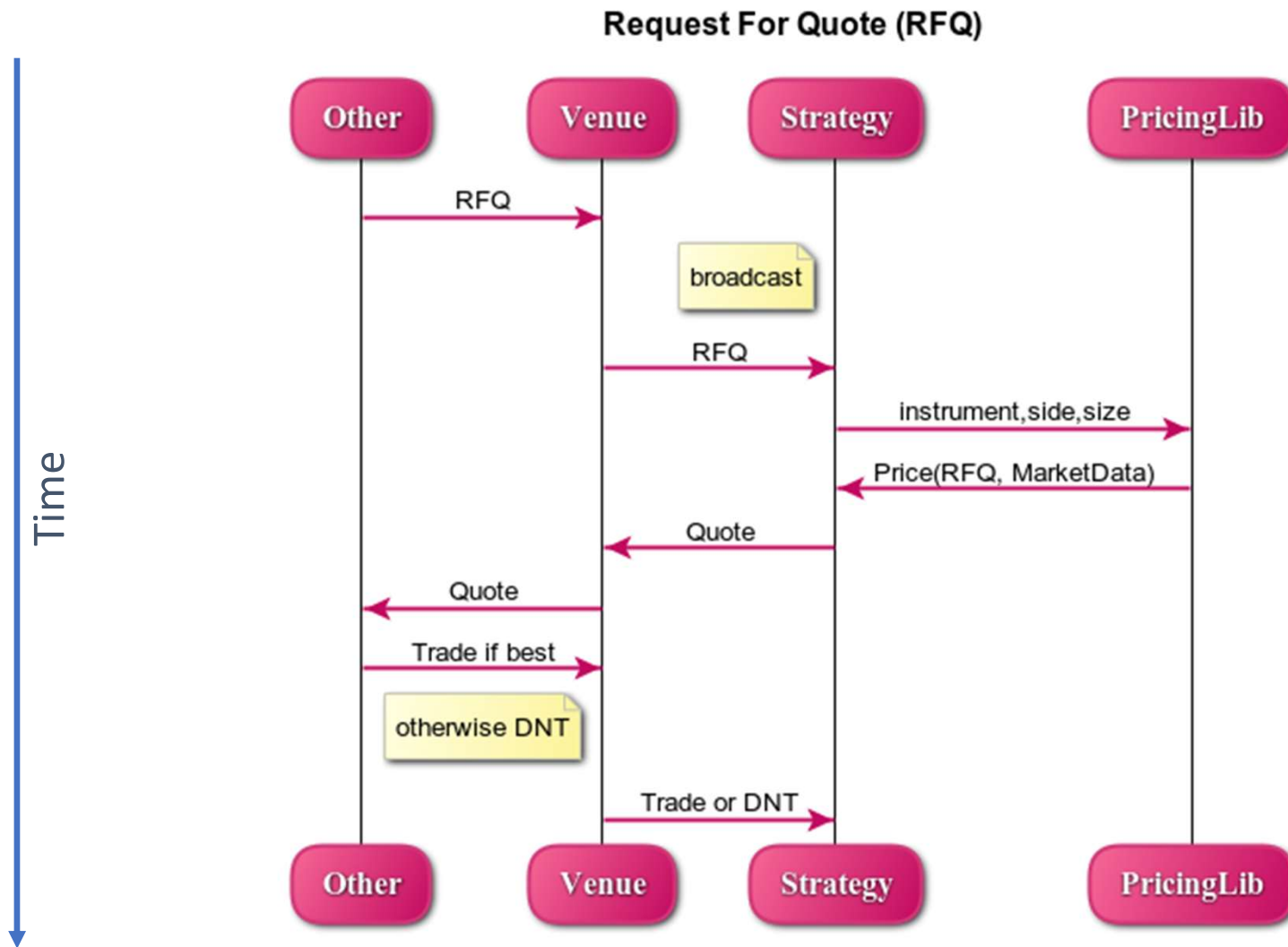
RFQ Attributes

- Symbol & Symbolology
 - AltSymbol
- Side
 - From whose perspective?
- Size
- PricingProtocol
 - Percentage, Yield, Spread
 - Benchmark if Spread
- HoldingBin | ASAP
 - Due Time
 - Firm Until
- RFQ ID
 - To be referenced
- [Sender]
- [# in competition]
- [PartialFillAllowed]

Venues and Data Vendors

VENUES		OrderBook	RFQ	Auction	Sweet spot	Data
MarketAxess Disclosed,OT			YES		>1M & <20K	RFQ, EvalPrice
Bloomberg ALLQ		YES	YES		[1 - 400]	Scraped
Bloomberg BOLT			YES			RFQ
ICE Bonds (TMC + Bondpoint)		YES	YES		[10 -200]	MD, RFQ
TradeWeb Direct		YES	YES		50-400	MD, RFQ
TradeWeb Institutional			YES			MD, RFQ, EvalPrice
MTS		YES			50K - 1M	MD
NYSE (ICE)		YES				0 MD?
Trumid				YES	6M	Mids, EvalPrice
Liquidnet				YES	3M	
GFI				YES	500+	
Chappy			YES		Muni	
JPM	Single Dealer		YES			MD
MS	Single Dealer	Quotes	YES			MD
Jane St	Single Dealer	Quotes	YES			
Headlands	Single Dealer	Quotes			Muni	
DATA		TYPE				
NEPTUNE		IOI				
TRACE		TRADE			Corp, Treasury, Mtge	
CP+		EVALUATED				
MSRB		TRADE			Muni	
SOLVE		IOI				
CC&S		MUNI IOI				
IDC		EVALUATED				

Sequence of events

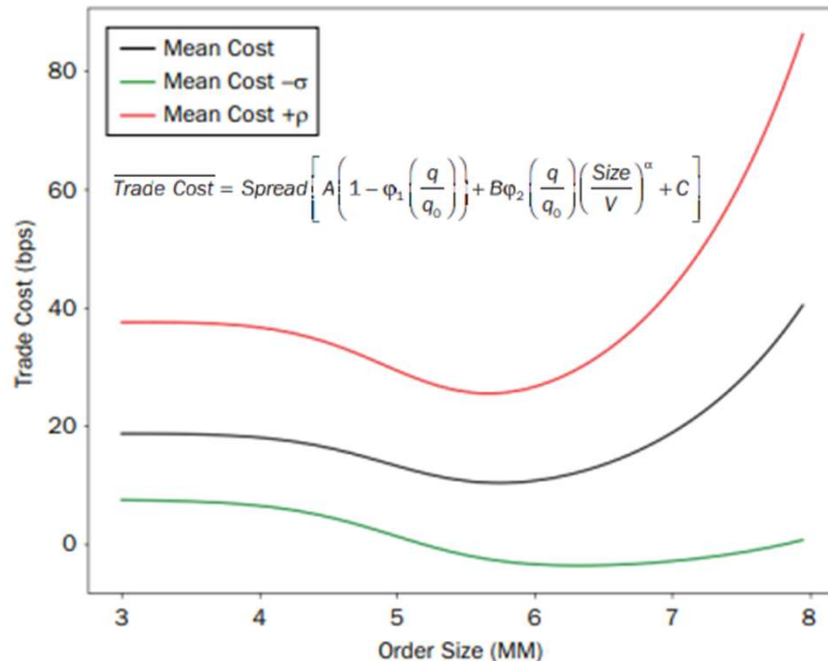


Trading Costs and Probability

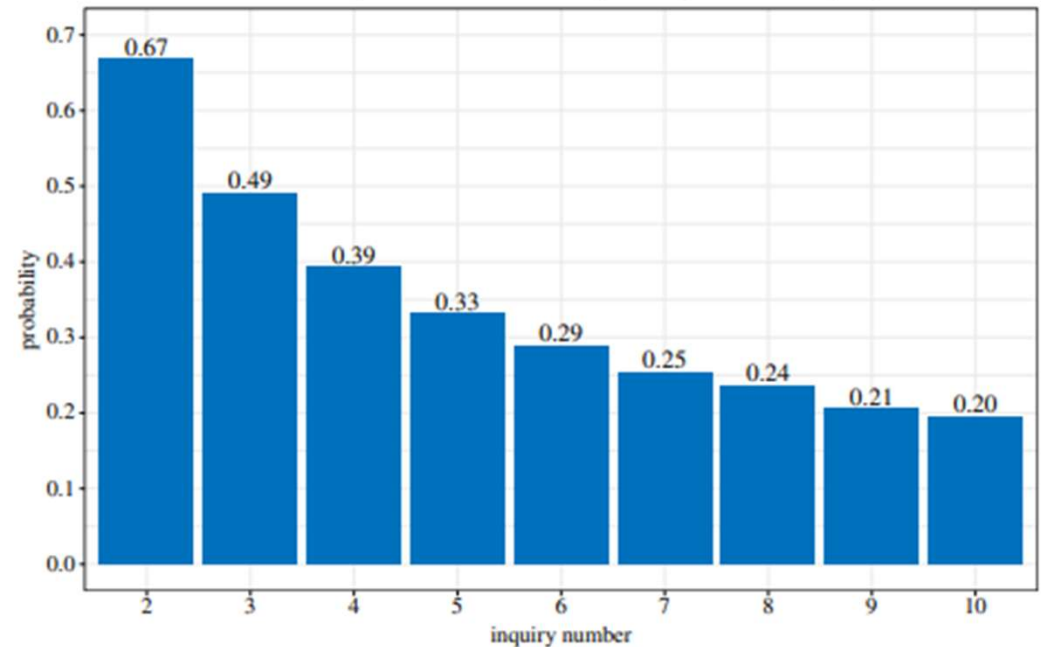
What is the optimal number of dealers to engage?

- What if dealers don't like competition (do not respond, or respond with a worse price)
- Are the dealers similar to each other?

Can I estimate the trading costs relative to a benchmark (CBBT, CP+, IBVL)



USD, Bid/Ask spread 50 bps (in the price space), amount outstanding 1 billion, time-to-maturity 5 years, age 2 years.

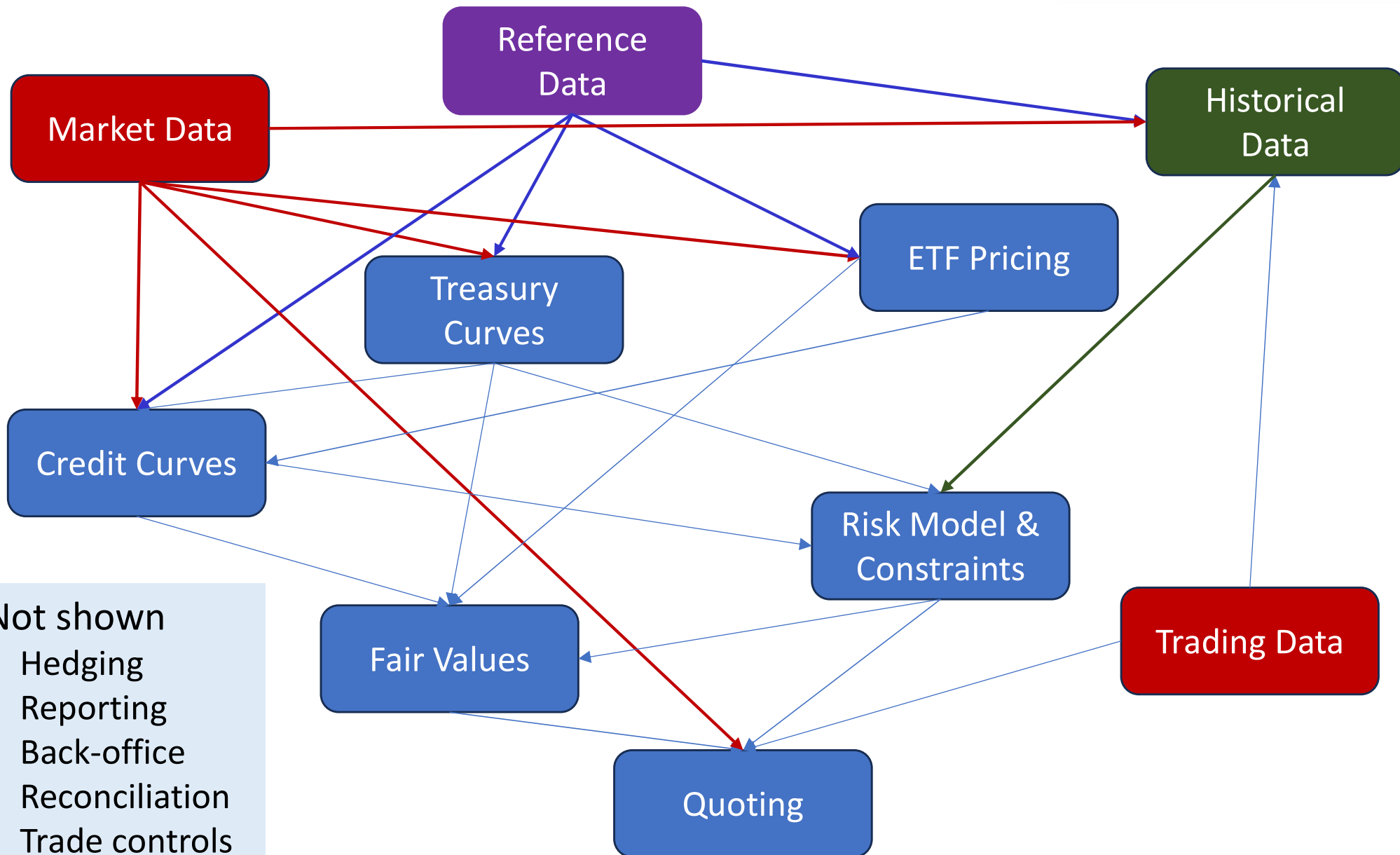


probability that a new dealer responds to a customer inquiry for a given inquiry number in child defined as a dealer responding to that inquiry for the first time during the entire child order.

A. Iogansen and V. Rashkovich https://assets.bbhub.io/professional/sites/10/1650256_TradingQuantReport.pdf

M. Kargar, B. Lester, S. Plante, and P. Weill, "Sequential Search for Corporate Bonds," NBER Working Paper 31904 (2023), <https://doi.org/10.3386/w31904>.

Incomplete* bond trading system



Systematic Bond Trading

❖ Preparing Static Data

- Acquisition (download, B-Pipe etc)
- Filtering/Normalizing, preprocessing
- Grouping

❖ Market Data Processing

- Maintain current state of data (order book)
 - Data from corporate bond venues
 - Treasury pricing
 - Trades (TRACE/MSRB/Venues)
 - Requests for Quotes
 - New feeds
- Fitting the Functional Form

❖ Curve Construction

- Updating reference curve
- Converting to z-Spread
- Fitting the Functional Form

❖ Quantifying The Opportunity

- Comparing bonds to the curve, evaluated pricing
- Comparing to pricing implied by ETFs

❖ Monetizing The Opportunity

- Use pricing in market/risk context
- Short term tactical signals
- Sending order/quotes/RFQ responses

❖ Hedging residual exposures

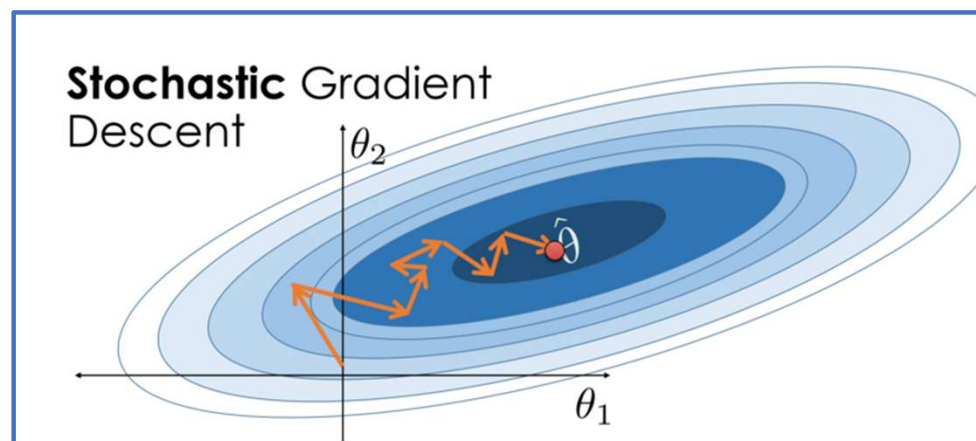
- Treasuries
- Treasury futures
- Fixed Income ETFs (e.g. LQD for corporates, TLT, SHY rates)

❖ Classical Portfolio optimization

- Implies ability to trade into desired portfolio
- Getting different prices is a correction
- Not getting into position is a correction
- Has a baked in assumption about liquidity

❖ Credit trading

- Multitude of correlated instruments
- Trading cost could be comparable to returns on shorter horizon
- Most instruments cannot be traded whenever you want
- Opportunity-driven optimization is one approach
- Portfolio Trading and ETF creation/redemption



https://optimization.cbe.cornell.edu/index.php?title=Stochastic_gradient_descent

Filtering Bonds - example

Filter Level (usage)	Basic (store in history)	Tradable (publish prices)	High Quality (used in calibration)
Min Amount Issued	\$50M	\$50M	\$250M
Min Amount Outstanding	\$40M	\$40M	\$200M
Max Average Price Spread	Any	\$3	\$2
Other Conditions	• Not defaulted • Not matured	• Not called (if callable) • Avg daily volume of large trades $\geq 250K$ • Quotes with tight bid-ask spread are present $> 40\%$ of the day	• Matures at least 1Y away • Avg yield spread < 20 bps • 4+ high quality bonds per curve • Junior/Senior attribution

Grouping Bonds

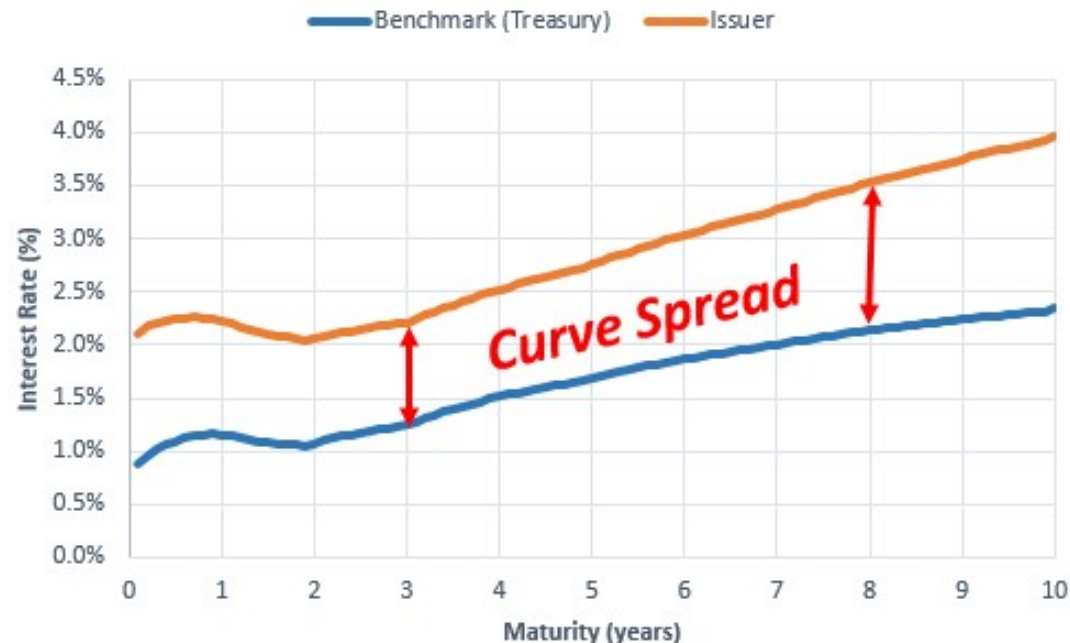
To construct the curves, we need to arrange the bonds into logical units, related by common features:

- Grouping Characteristics
 - Issuer (e.g., AAPL, GS, MO)
 - Industry (e.g., Financials, Industrials)
 - Rating (e.g., A-, BB+)
 - Industry & Rating (e.g. Financials/BBB)
- Features
 - Distinguish between junior and senior bonds, if necessary
 - Spread from a relevant benchmark curve (called the reference curve)
 - Curve captures common risk of the entire group



Converting Prices to Spreads

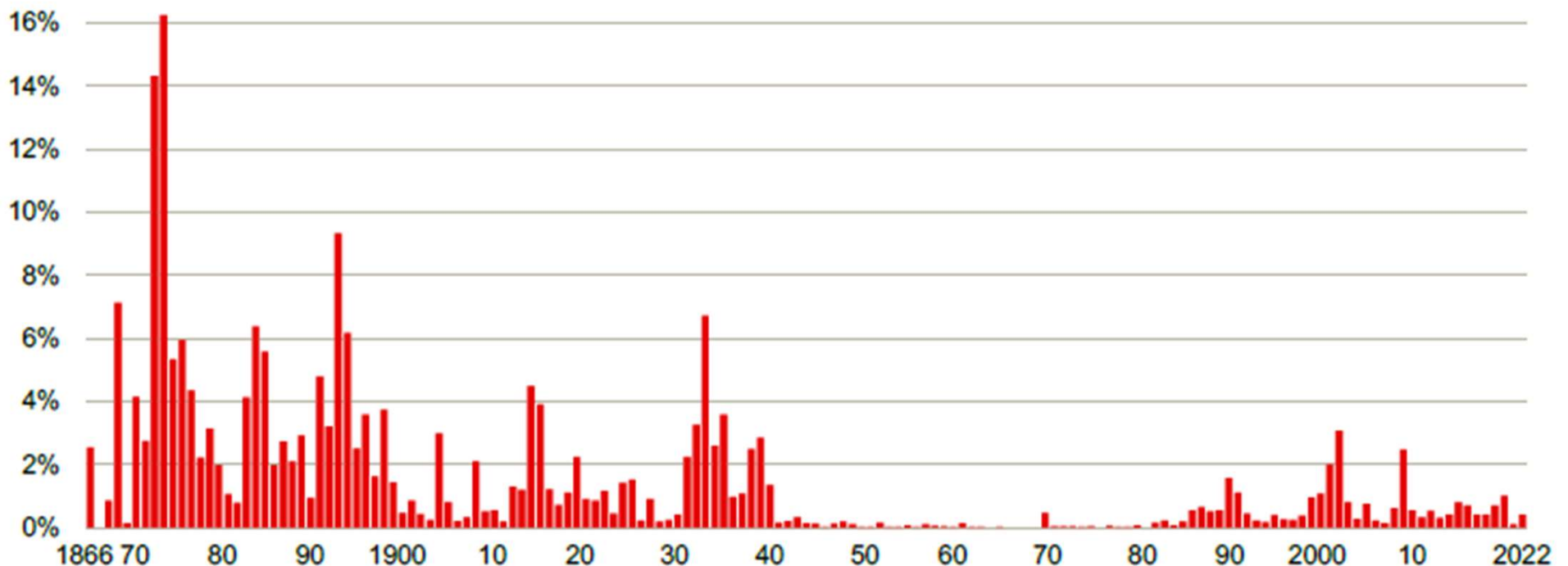
- Initialization
 - Prepare the structures
 - ✓ Construct cash flows, calendars, etc.
 - ✓ Construct bond objects
- Real-Time Processing
 - Obtain market prices
 - Calibrate the reference curves
 - Calculate the spreads for each bond
 - ✓ Imply yields from clean prices
 - ✓ Calculate the corresponding values from the base curve
 - ✓ Compute the spreads



Factor Investing in Credit

- Default rates cluster in time with a serial correlation of 0.63
- Average US credit premium $\sim 1.80\%$ after 1936 (1.35% IG and 3.35% HY)
- Average default rate 1.2% (0.1% for IG and 2.9% for HY)
- Annual loss rate is 1.0% (0.2% for IG and 2.7% for HY)

Percent of total value defaulting



Source: Giesecke, Longstaff, Schaefer and Strebulaev (2011), updated to 2022 by Dimson, Marsh and Staunton using Moody's data.

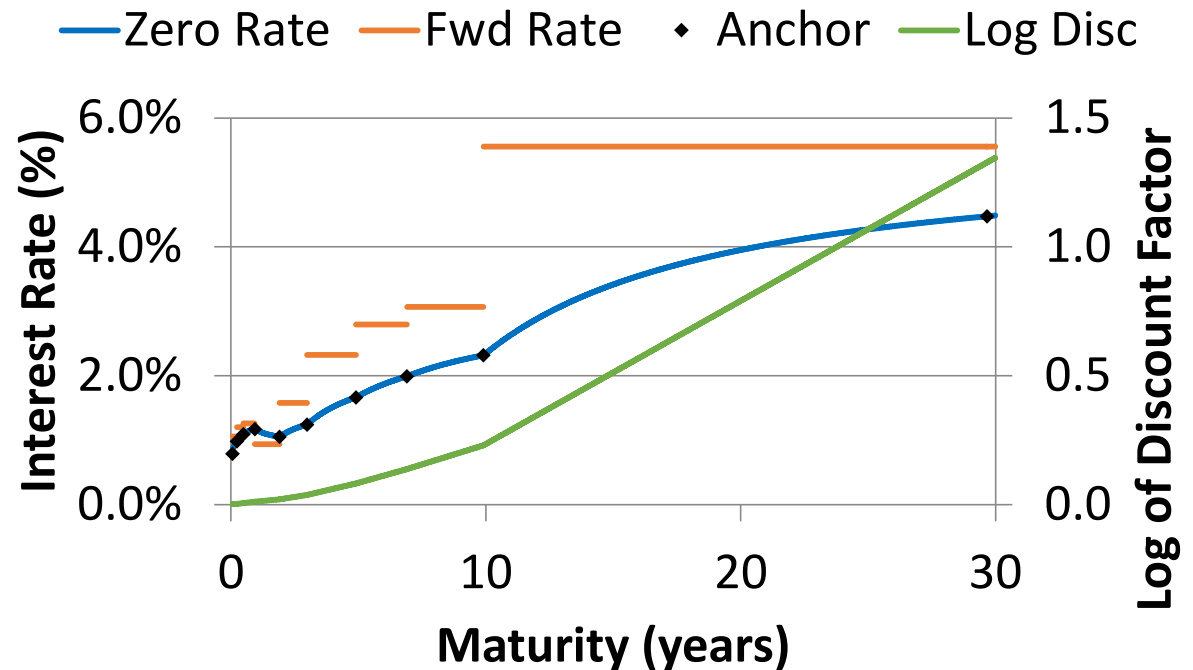
Buy-sell asymmetry

- Most bonds not shorted and are held to maturity
 - munis are a good example
 - Corporates are expensive to short
- Buyer buys **a** bond (any in a group)
 - Chooses the best price among many nearly fungible bonds
- Seller can be motivated (he needs to close on a house, a yacht)
- Seller sells **the** bond (what he owns)
 - Seller sends bid-wanted (RFQ)
 - Dealers compete to buy “at bid”
 - Dealer turns around post liquidity on exchange
- May vary quoting logic between bidding and offering

Constructing the Treasury Curve

- Arrange instruments (anchors) by maturity
- Bootstrap the zero-coupon rates for each anchor
 - Find the (unique) zero-coupon rate to match the theoretical price of the anchor to its market price
 - Interpolate using
 - ✓ constant forward or
 - ✓ linear log of discount

Treasury Curve Bootstrapping



https://www.atlantafed.org/-/media/documents/research/publications/economic-review/2004/vol89no3_fisher.pdf
<http://gouthamanbalaraman.com/blog/quantlib-term-structure-bootstrap-yield-curve.html>



Nelson-Siegel 4-Parameter Model

- Assumes implied forward in a functional form:

$$f_t = L + Se^{-tD} + CDte^{-tD} = L + e^{-tD}(S + CDt)$$

- Resulting zero curve is then

$$z_t = \frac{1}{t} \int_0^t f(s) ds = L + (S + C) \frac{1 - e^{-tD}}{tD} - Ce^{-tD}$$

- $z_t(0)$ and $z_t(\infty)$?

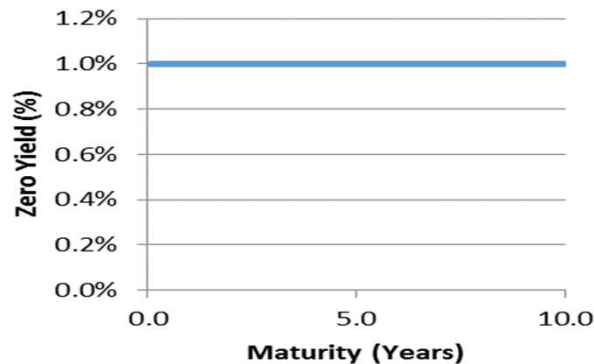
Nelson, C., & Siegel, A. F. (1987). Parsimonious modeling of yield curves. The Journal of Business, 60(4), 473-89.

Constructing the Issuer Curve – Intuition

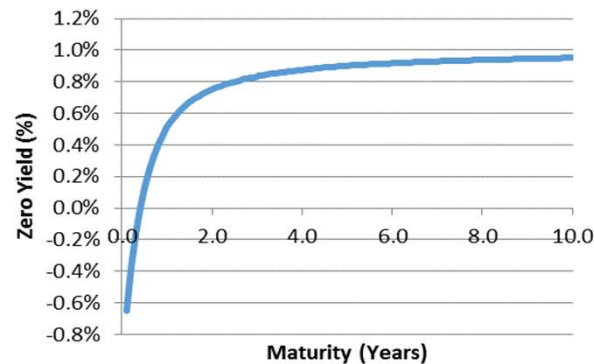
Geometric Interpretation of Parameters

- Level (z_∞), slope ($z_0 - z_\infty$), curvature and decay D
- L , $(S + C)$ and $-C$ correspond to the top 3 principal component factors
- D determines the shape of the curve (location of hump/bend)
- One can use principal components derived from the market data

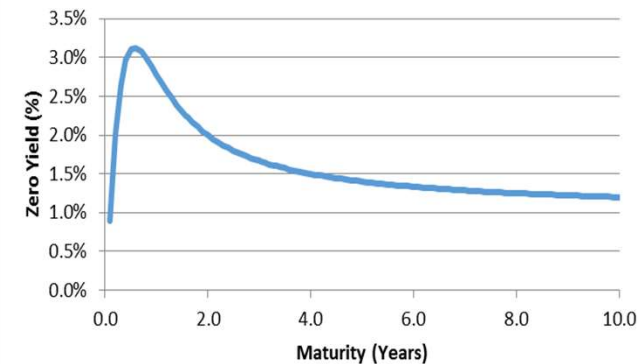
NS Model: $L=1\%$, $D=4$



NS Model: $L=1\%$, $S=-2\%$, $D=4$



NS Model: $L=1\%$, $S=-2\%$, $C=0.1$, $D=4$



Constructing the Issuer Curve

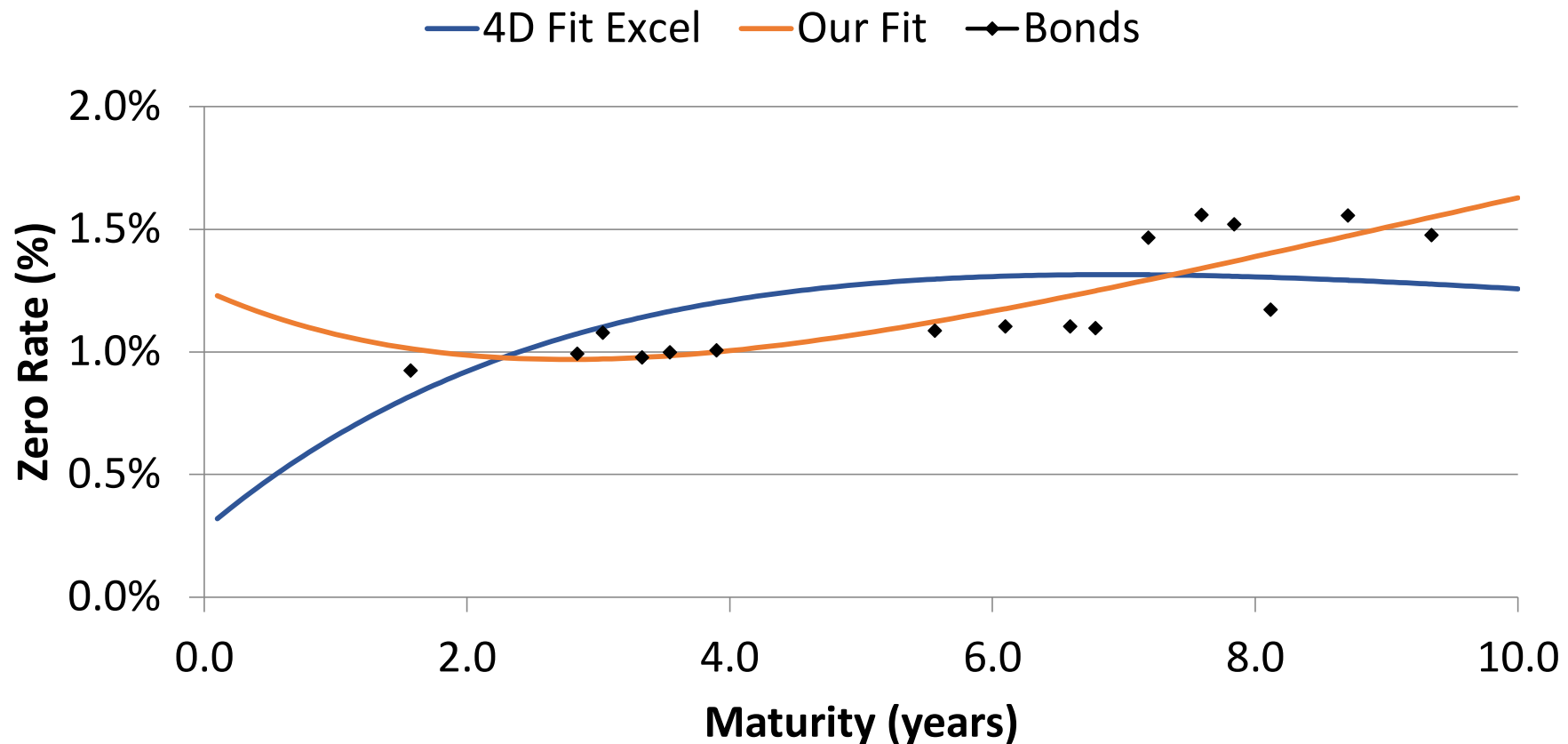
Optimize model parameter choice

- Exact non-linear optimization in 4-D is computationally intensive
 - You inputs are prices or yields
 - Curves describe forward to zero rates
 - You need to move between the two in order to optimize
- Alternative formulation allows fast progress
 - Work with z-Spread, not with prices directly
 - Finding just level, spread and curvature is a linear least squares problem
 - Decay determined by 1D numerical minimization of the weighted square error



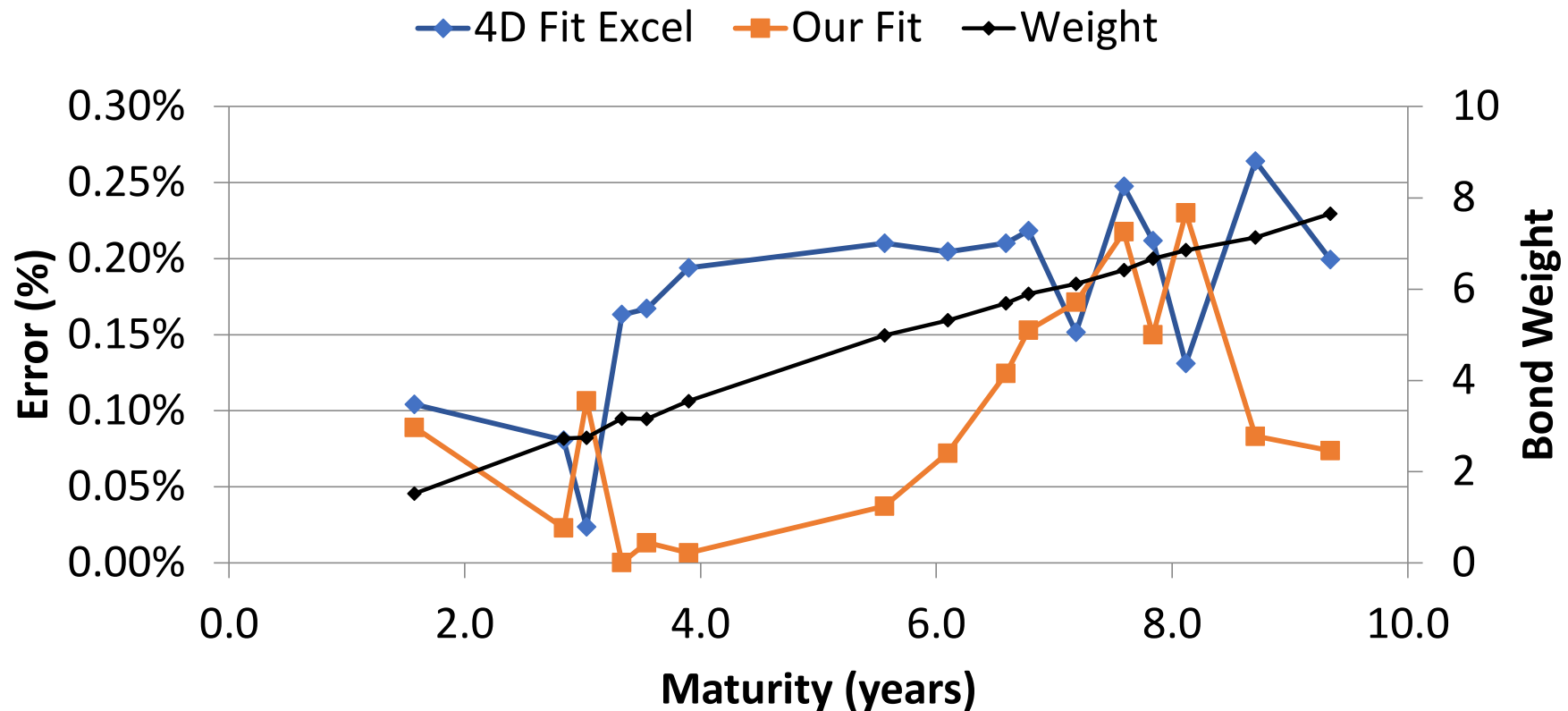
Fitting the Issuer Spread Curve

BAC 2017-06-01 08:13:38: Raw 4D Fit vs. new scheme



Issuer Credit Spread – Fitting Error

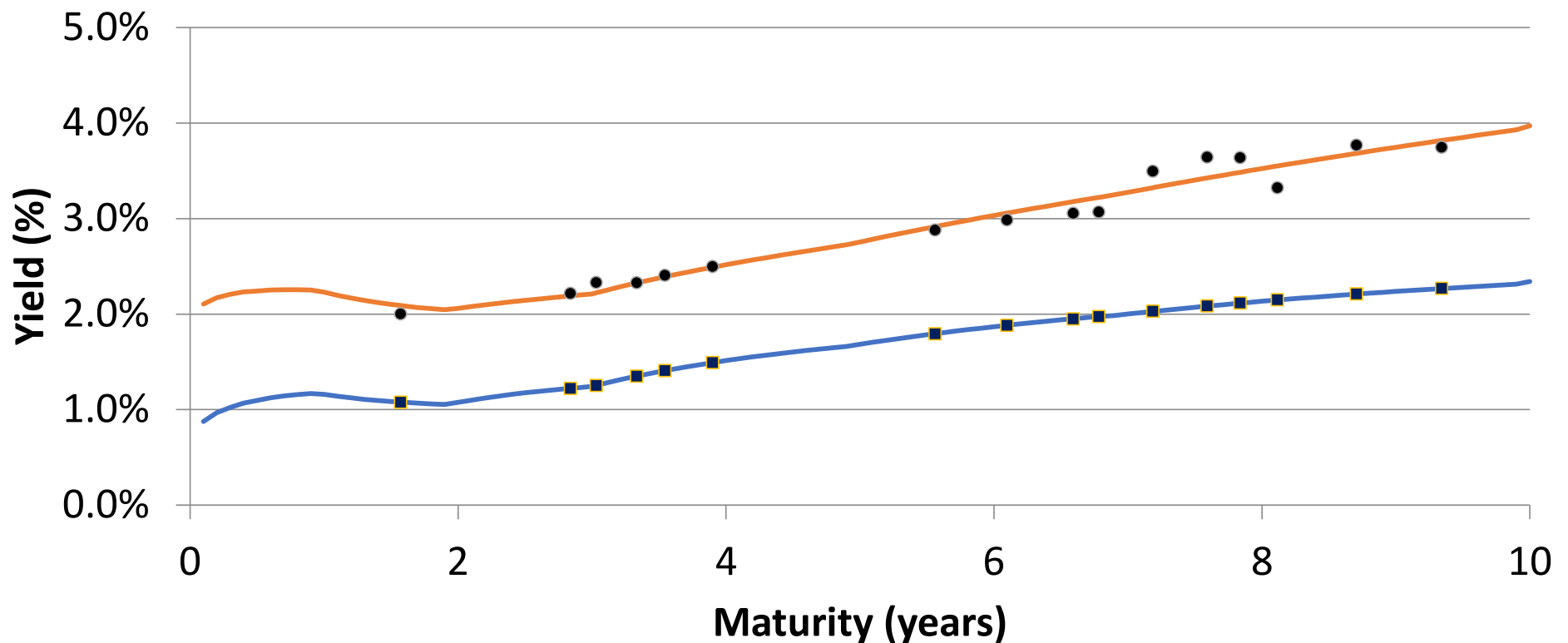
BAC 2017-06-01 08:13:38: Fitting Error



Constructing the Issuer Curve – Final Fit

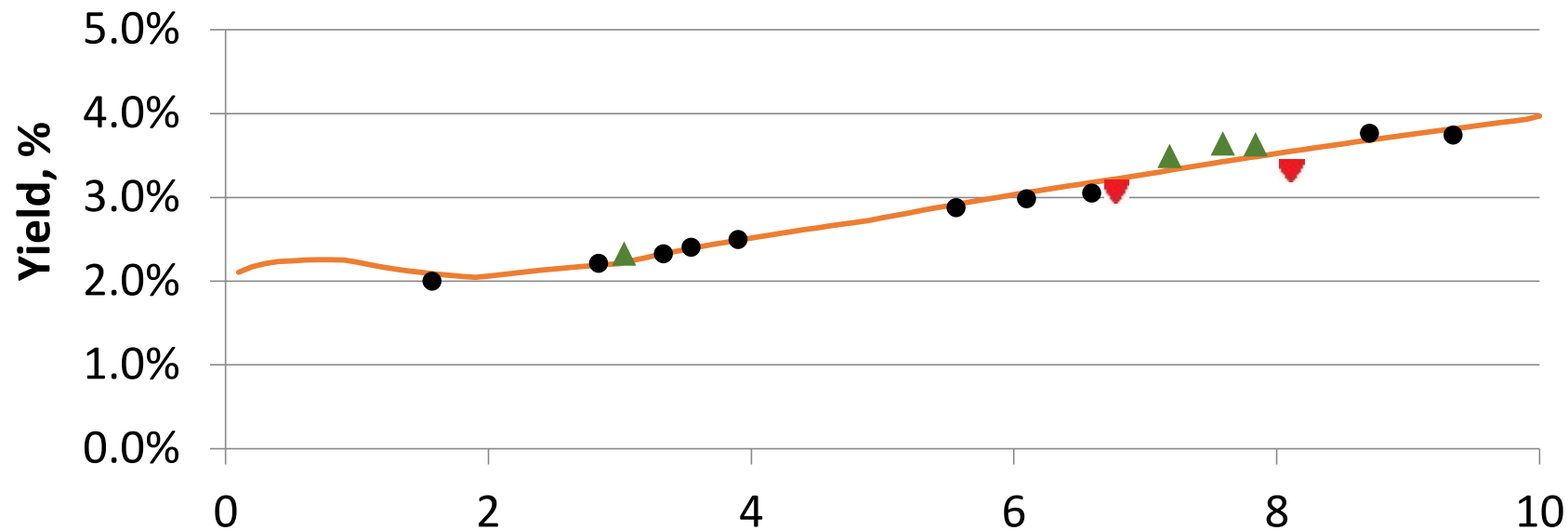
2017-06-01 08:13:38 – BAC vs. Treasury

— Treasury — BAC • Bonds ■ BaseLevel



2017-06-01 08:13:38 BAC – CURVE

— BAC ▲ Cheap Bonds ▼ Rich Bonds ● Inline Bonds



Notes on Curve fitting

- Some bond prices are more reliable than others
 - Volume of trades [issue size, age, index inclusion]
 - Volume of market data
 - Bid-offer spread
 - Need enough bonds to build a curve (>4 for Nelson-Siegel)
 - Simplicity of bonds priced off the yield curve
 - Bullet [no options, fixed coupon]
 - Callable/Puttable => yield to worst, convention
 - Senior/Junior → need to build Junior curve
 - Issued by a subsidiary with a different rating → mispriced
 - 144A / Reg S → less liquid, size constraints
 - Insured or asset-backed → will always look rich
 - Step-up or step-down coupon → different cash flow

Homework assignment

1. Choose from WRDS download files(s) enhanced end of month data for 4+ of your favorite large issuers(e.g. MSFT, AAPL) for a period of year+:
 - <https://wrds-www.wharton.upenn.edu/pages/get-data/wrds-bond-returns/wrds-bond-returns/>
 - Make sure your issuers have enough bonds to cover maturity spectrum.
 - Dataset covers up to 2025, so choose longer periods
2. Filter data: `SECURITY_LEVEL` and `RATING` to have consistent values across bonds
3. Plot Treasury spread `T_Spread` vs `DURATION` for each issuer
4. Use basis of your choice to fit the spread curve with a 3 parameter model of YOUR CHOICE
 - E.g. $\text{spread} = a + b \cdot \log(1 + \text{TMT}) + c \cdot \text{TMT} + d \cdot \text{coupon}$, as shown in the example below
 - TMT – Time-To-Maturity. Duration could be a better options
 - Consider weighting regression on reliable points (actively traded, large issues)
 - take a look at outliers: is there anything systematically special about these bonds?
5. Built a duration-neutral portfolio (short rich bonds, long cheap bonds)
 - Duration-neutral portfolio: $\text{sum}(\text{position} \cdot \text{PercentagePrice} \cdot \text{Duration}) \sim 0$
 - Monthly rebalance as data allows
6. Evaluate its return, PNL, turnover etc on the next month data.
 - Make necessary assumptions.
 - make sure it includes accruals (or add them yourself as a 1/12 of a coupon)
 - Summarize your findings

Can you tell a good curve from bad curve?

- What is the purpose?
- Back-test “sanity check” for mean reversion
 - Buy cheap at mid
 - Short rich at mid
 - Control risk (e.g. DV01-neutral)
 - Compute PNL, analyze “losers”
- Back test for real
 - Replay [possibly preprocessed] market data
 - Compute your pricing model
 - Replicate your actions (place orders, RFQ responses)
 - Develop a “fill model” (simulated counterparties)
 - Compute PNL, analyze exposures, “losers”
 - Update your model

Risk model (one of)

Consider a portfolio of bonds, with positions w_i and prices P_i , $i=1..N$

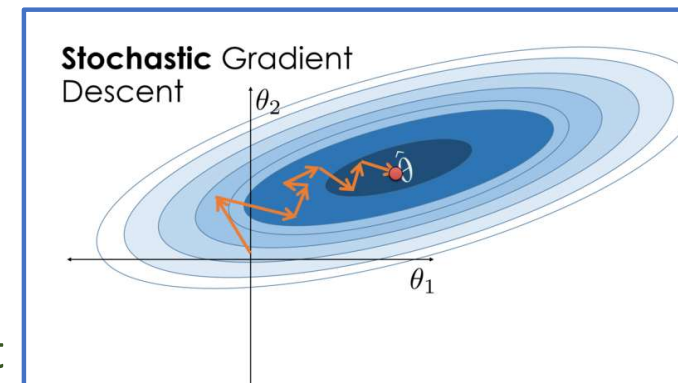
- Note that trading ultimately done in dollar terms.
- $P = w_i P_i$ with double indexing representing summation

Suppose we have a signal α_i and utility function U

- $U = w_i \alpha_i - \lambda R(\mathbf{w})$ where R is an expected variance of portfolio PNL.

Suppose we can model each bond via a combination of real-valued factors f_j , $i=1..M$

- $P_i = P_i(f_1, f_2, \dots)$
- In our case factors can be
 - Treasury curve level, slope, curvature
 - Principal components of treasury rates
 - Issuer spread curve level, or other parameterizations
- $\Delta P_i = \sum \frac{\partial P_i}{\partial f_j} \Delta f_j$ summation implied by repeating subscript



Variance of the portfolio is then [using double index for summation]

- $R = \langle (\Delta P)^2 \rangle = \left\langle \left(w_i \frac{\partial P_i}{\partial f_j} \Delta f_j \right) \left(w_k \frac{\partial P_k}{\partial f_m} \Delta f_m \right) \right\rangle = w_i \frac{\partial P_i}{\partial f_j} \langle \Delta f_j \Delta f_m \rangle \frac{\partial P_k}{\partial f_m} w_k$
- $U = \alpha^T \mathbf{w} - \lambda \mathbf{w}^T \mathbf{L}^T \mathbf{C} \mathbf{L} \mathbf{w}$
- $L[N \times M]$ loadings aka sensitivities extracted from yield curves
- $C[M \times M]$ covariance matrix estimated e.g. from time history of curve parameters

Portfolio Trading and ETFs

- Fixed income ETFs volume is comparable with trading volume of underlying
- Portfolio trading grew to 10% of credit volume
 - Can be tracked by TRACE timestamps historically
 - PT flag added to TRACE May 2024
 - Create noise in individual bond pricing
- Convenience of a single price for buy-side
- “Time is alpha” but is price really better than buying piece-meal?
- PT seems to be cost-efficient in the US, not a given in EUR, GBP

Fixed income ETF AUM \$1.5 Trillion

[Vanguard Total Bond Market ETF BND](#) with \$94.94B in assets

<https://www.etf.com/topics/fixed-income>

Exchange-Traded Funds (ETF)

- Trade as equities
- Much more liquid than individual instruments
- Prices move with supply-demand (equity side) and with underlying securities (Fixed income side)
- Price divergence is ~~removed~~ limited by creation-redemption mechanism
 - Only executed via Authorized Participants (AP)
 - Basket are negotiable
 - Each ETF has some peculiarities
- Major managers: Blackrock (iShares), Vanguard, State Street, major banks
- Great for:
 - Getting broad exposure to a segment
 - Hedging residual risk (rates and credit)
- Not to be confused with “Exchange Funds” where investors pool their individual portfolios

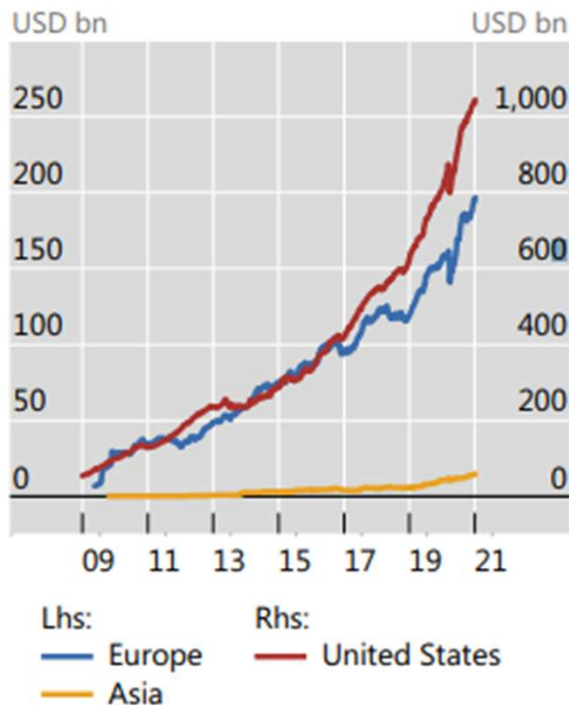
Liquidity and Bond ETFs

- Bond's minimum trade size/increment \$1000
- Major brokerage minimum size \$100,000
- Many bonds not accessible to retail investors
- Trading cost for 1-10 bond is 3x that for 1000 bonds
- Per trade ticket charges don't favor many small trades
- Access to illiquid market benefits from introduction of basket trading opportunity according to [Market Accessibility, Corporate Bond ETFs, and Liquidity](#), C.W. Holden, J. Nam (2019)
 - Used TRACE data
 - out-of-index bonds to benchmark
 - improvement for 144a, high yield, large tenors

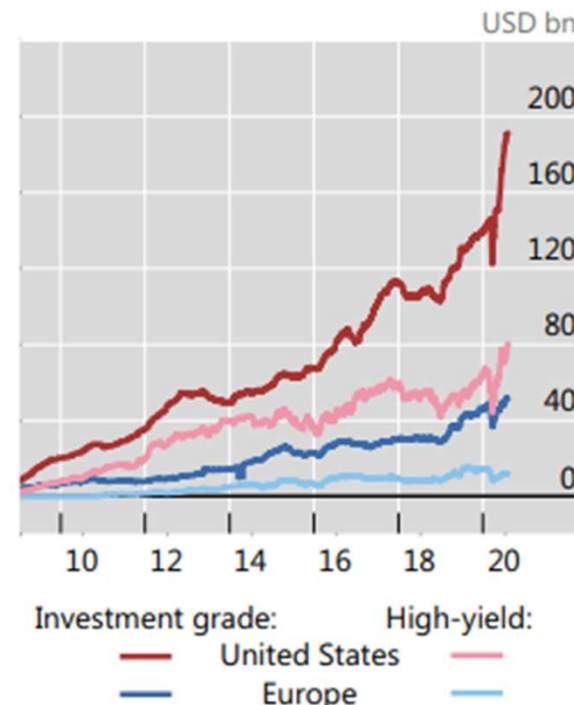
Bond ETF Arbitrage

- Creation redemption is ~10% of ETF trading volume
- Largest AP's:
 - Banks: BAML, GS, JPM, MS
 - Market makers: Citadel, Jane St, Flow Traders, Virtu

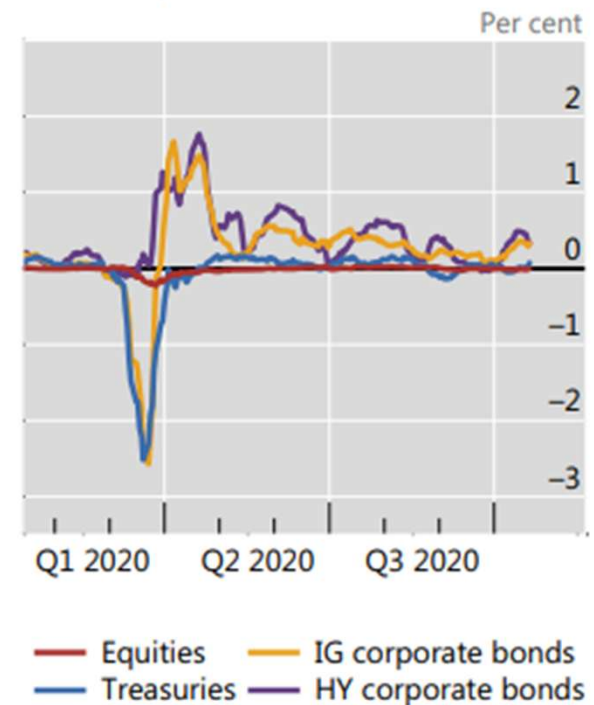
AUM rose globally¹



Growth of US corporate bond ETFs accelerated after March 2020²

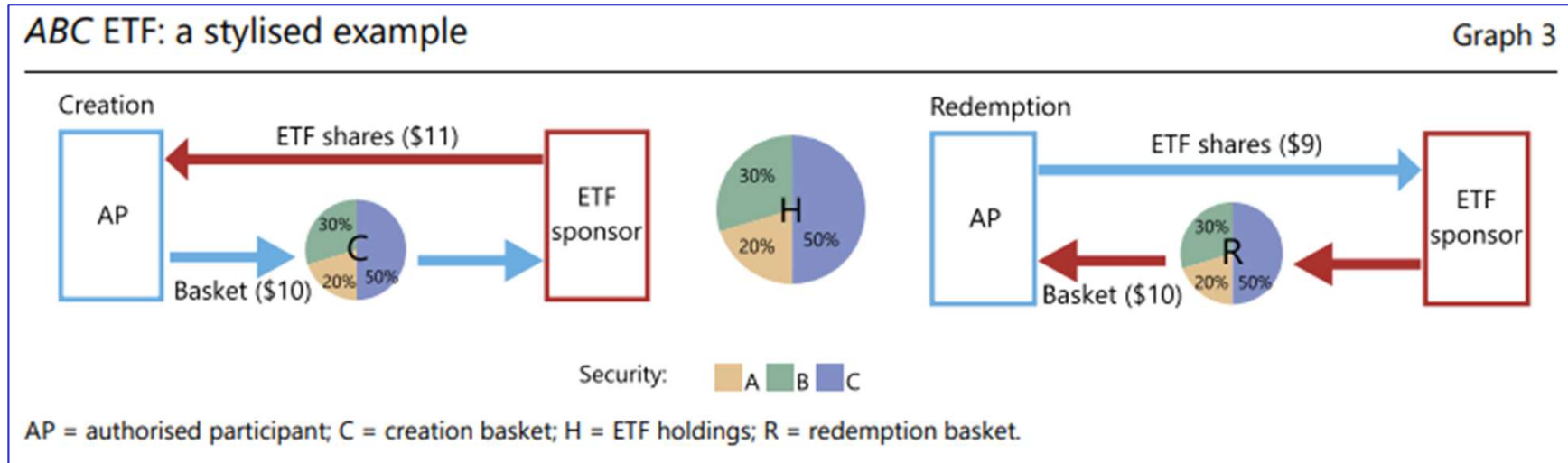


ETF premiums swung sharply in March–April 2020³



[The anatomy of bond ETF arbitrage](#) by K. Todorov (BIS Quarterly Review, March 2021)

Bond ETF Arb Illustration



Suppose ETF trades at premium:

- $NAV@T-1 = \$10 (3A + 2B + 5C)$
- ABC ETF price is **\$11**

AP Creation Arbitrage

- Buy A, B, C in secondary market (-\$10)
- Exchange {A,B,C} for shares of ETF (~\$0)
- Sell ETF on exchange (+\$11)

Suppose ETF trades at discount:

- $NAV@T-1 = \$10 (3A + 2B + 5C)$
- ABC ETF price is **\$9**

AP Redemption Arbitrage

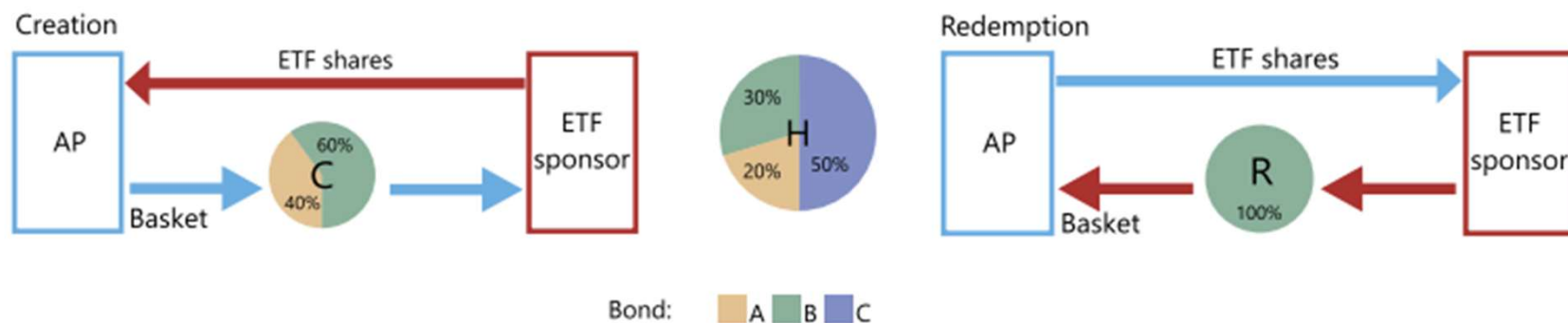
- Buy ETF on exchange (-\$9)
- Exchange shares of ETF for {A,B,C} (~\$0)
- Sell A, B, C in secondary market (+\$10)

[The anatomy of bond ETF arbitrage](#) by K. Todorov (BIS Quarterly Review, March 2021)

Bond ETF Arb Illustration II

ABC ETF: a stylised example with bonds

Graph 4



AP = authorised participant; C = creation basket; H = ETF holdings; R = redemption basket.

[The anatomy of bond ETF arbitrage](#) by K. Todorov (BIS Quarterly Review, March 2021)

For illiquid bonds 1000+ names in benchmarks, baskets are impossible to replicate:

- C/R baskets cover by 50-200 names, subject to negotiation
 - ETF managers want to get rid of maturing/aging bonds
 - ETF managers like illiquid bonds to reduce tracking errors
- NAV methodology may be proprietary to exchange find (reverse engineer?)
- Creating a good replicating basket with a subset of bonds **is a non-trivial problem**
- Other participants can engage AP to create/redeem for them. Unit size ~100K shares

Timing risk

- Premiums/discounts may disappear
- NAVs for creation/redemption are not known at time of trade

Portfolio Trading in Fixed Income

- Single price paid for a portfolio of bonds
- Price can be established in competition over RFQ protocol
- Portfolio trading grew from ~0 in 2017 to 10%-15% of TRACE volume
- PT trades are flagged as such in TRACE since May 2023
- [Portfolio Trading in Corporate Bond Markets](#) by M. Jeffery and Z. Torodova (2023)
 - 40% improvement in trading costs relative to piece-meal RFQ'ing (?!)
 - Mostly benefits illiquid bonds
 - Explained as a spillover of ETF creation/redemption activity
 - Most pronounced for buys when ETF trades cheap, and sells when ETF rich.
 - Dropped considerably since 2023
- Trading venues such as Bloomberg, MarketAxess, Tradeweb, Trumid rush to add portfolio trading functionality to their workflows and trading stations
- Individual bond prices still need to be determined correctly for reporting and allocations to sub-accounts.
 - Not a trivial problem – there is not true “single price” for a PT

Fixed Income Vs Equities Trivia II

Measure	Fixed Income (Municipal)	Fixed Income (Corp)	Equities
Market Size,\$	4.0T	11.5T	47T
Issuance,\$	400B	2.3T	400B
Daily Trading Volume,\$	10B	46B	481B
Electronic Volume,\$	1B	3B (*)	350B
Number of trades	45k	70k	59m
Electronic Trades	4k	12.5k	50m
Instruments	1MM	100K	10K
Average trade size,\$	250K	650K	8K
Electronic trade size,\$	200K	240K	7K

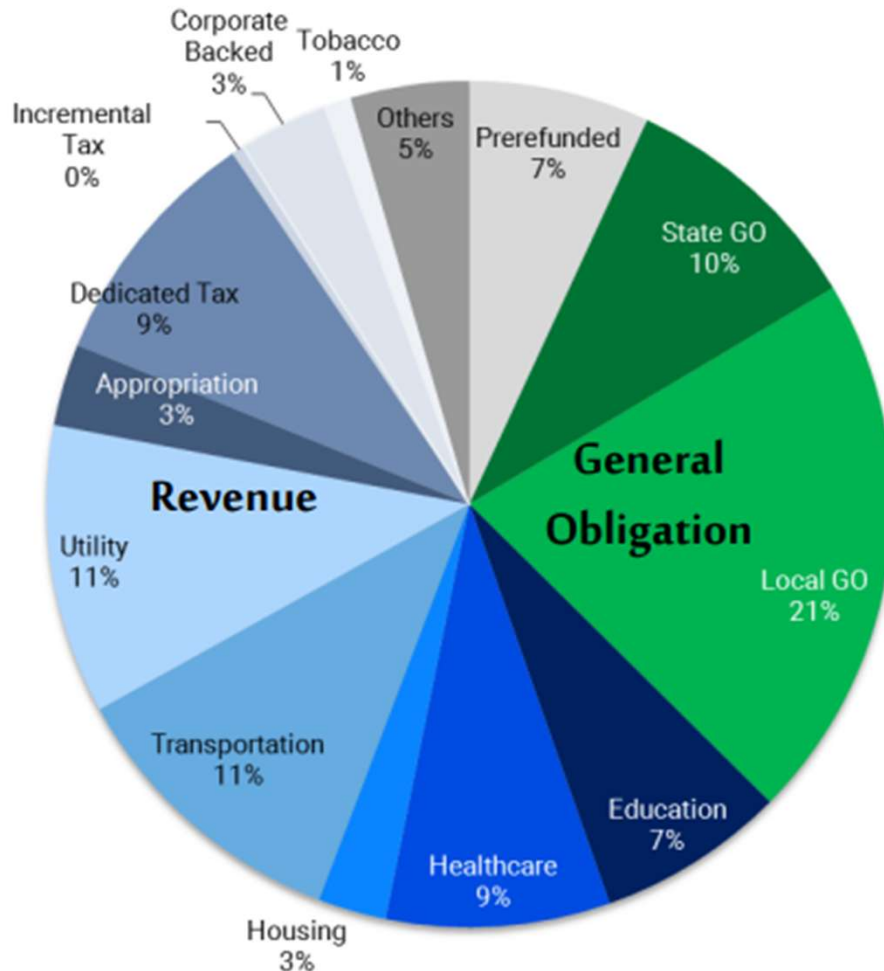
*Sources: FINRA, SIFMA, MSRB

Munis – a new FI darling

- 1MM unique municipal bonds
- Very low historical default rate
- Pre-refunded (no new issuance), insured*, or asset backed
- Mostly exempt from local and federal tax
- Mostly illiquid.
- Cannot be shorted.
- Marked against a MMD muni curve (or BBG equivalent)
- Mostly held by individual investors
- Sold when money is needed (retirement, big purchases, inheritance)
- Trades reported by dealers on <https://www.msrb.org/> (much like TRACE)

* Investor attention and municipal bond returns, by K. Cornaggia, J. Hund, G. Nguyen

Munis by security and issuer type



General Obligation (30%)

- Backed by “full faith and credit” of US municipalities with taxation authority
- States, cities, counties, school districts, etc.

Revenue (63%)

- Secured by revenue pledges
- Public enterprises: utilities, toll roads, airports, etc.
- Nonprofit organizations: hospitals, universities, etc.

Pre-refunded / Escrowed to Maturity (7%)

- Funds securing bonds invested in indirect or direct federal obligations and placed in escrow

- 15% Federally Taxable
- 78% Federally Tax-Exempt

- 75% have coupon in (5,6] %
- 10% have coupon in (4,5] %

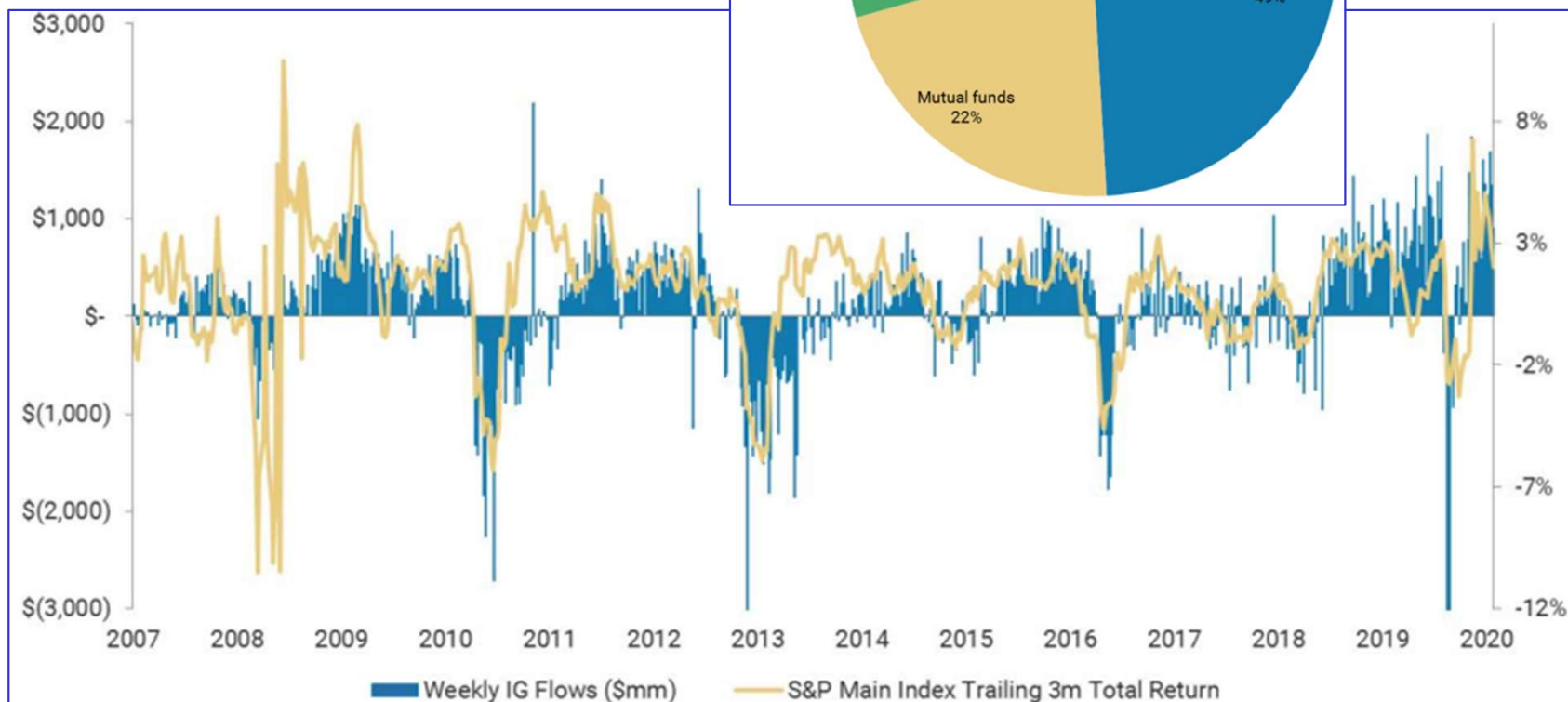
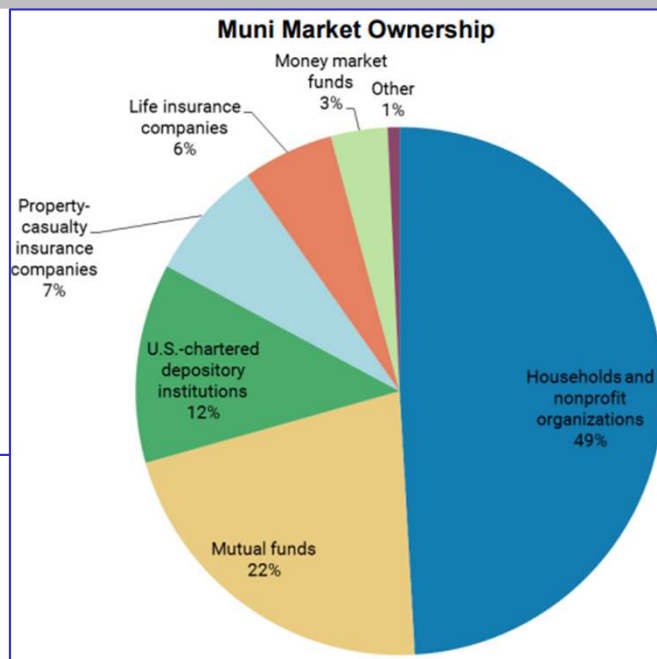
- California
- New York
- Massachusetts
- Illinois
- **Others**

Sources: <https://www.sec.gov/news/studies/2012/munireport073112.pdf> and Morgan Stanley Research, S&P Dow Jones Indices, Bloomberg as of Aug 2020

Market dominated by households.

Fund Flows Tend to Track Total Returns

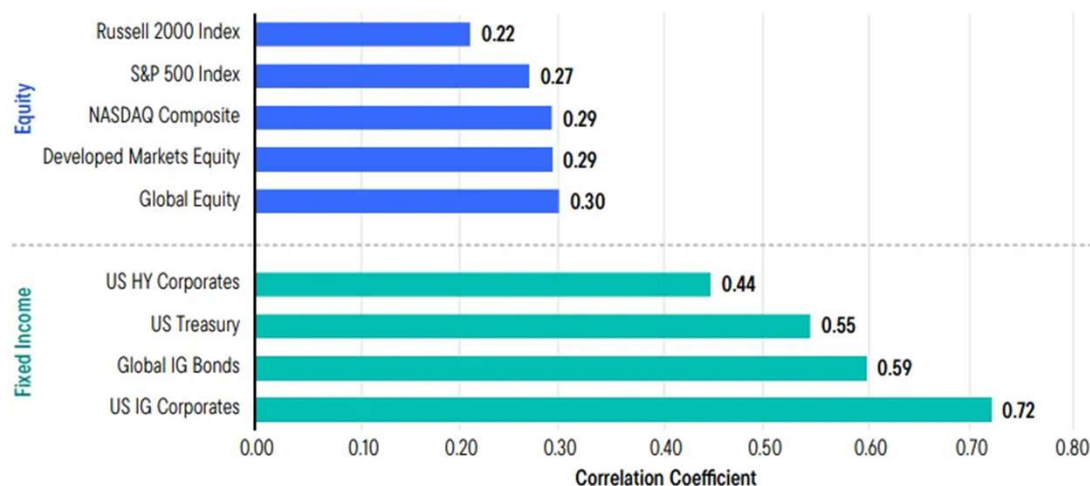
- Muni buyer dynamics create conditions for recurring demand & momentum cycle.
- Individual investors exercise an outsized influence on the market.



Source: Morgan Stanley Research, S&P Dow Jones Indices, EPFR

Correlation to treasuries and corporates: on-on-off

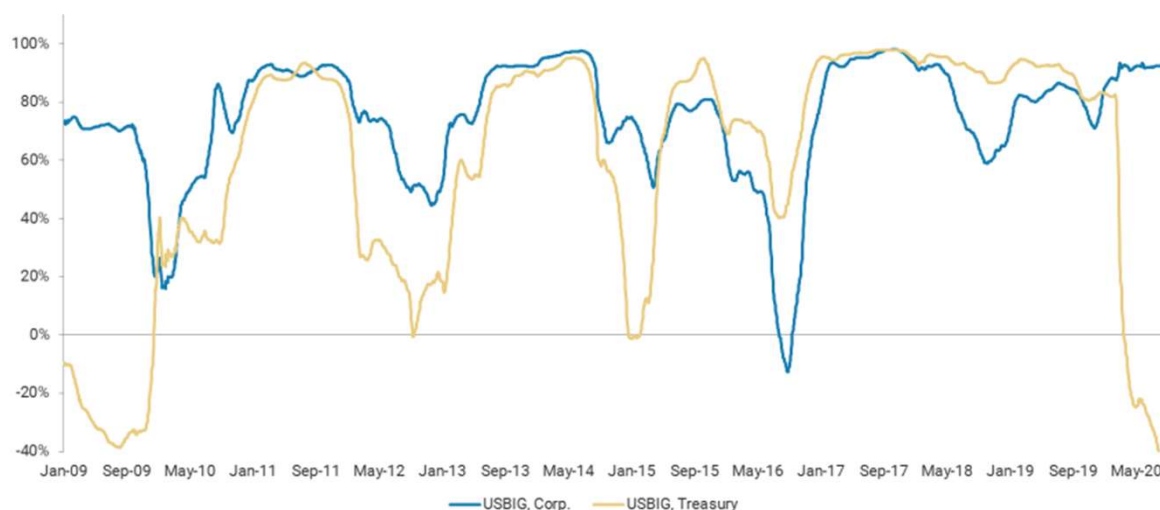
Correlation of Municipal Bond Index to Equity and Other Fixed Income Asset Classes
Last 20 Years Ending 11/30/2025



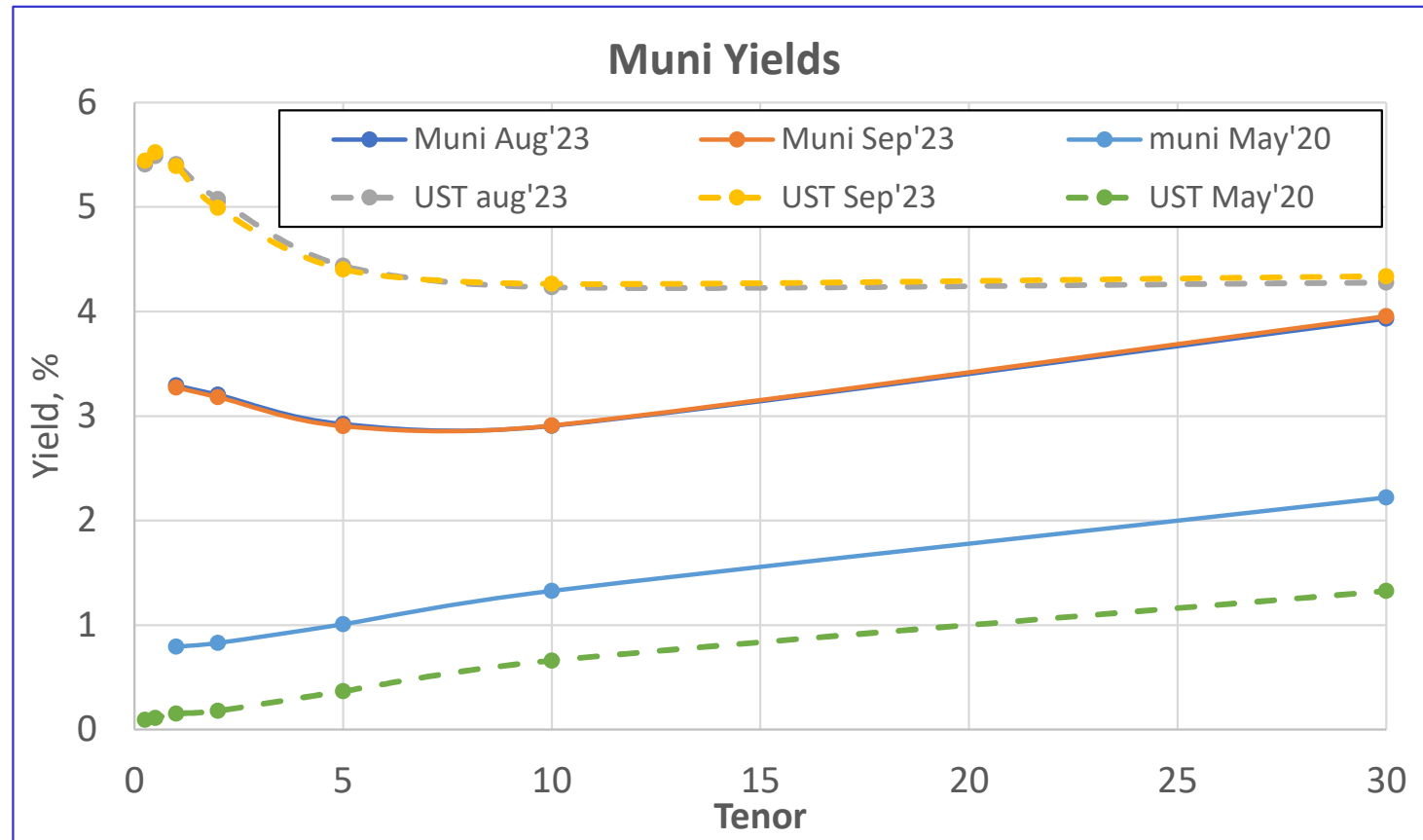
- Munis are a good diversifier
- Long-running correlation are misleading
- TEY – Tax-Equivalent Yield
- Gained popularity for tax-harvesting

- Long-running correlation are misleading
- Not sure about current returns
- Bonus to those who find data for last 5 years and extends this chart

Trailing 12m correlation of rolling 3m returns: Munis versus Treasuries, corporates



Municipal Bonds: negative spread



[What Makes the Municipal Yield Curve Rise?](#) by A. Kalotay and M. Dorigan, 2008

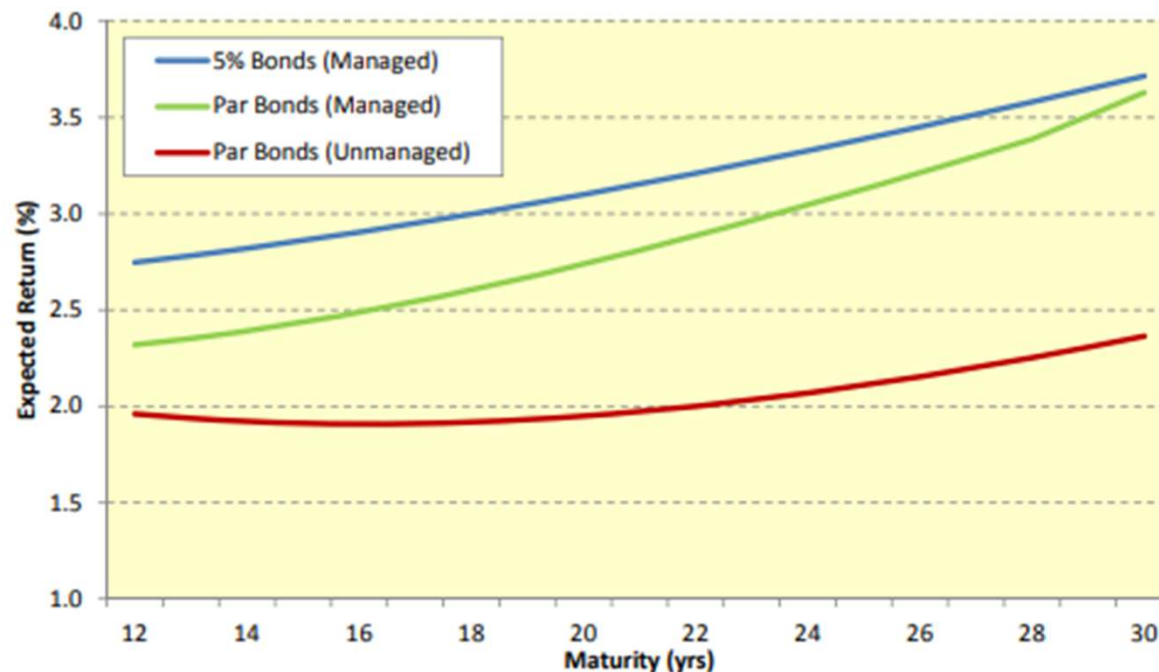
We have shown that as long as the yields are based on callable bonds, the curve will always be upward sloping, regardless of the shape of the underlying optionless curve. The optionless yields implied by the callable yields can be obtained iteratively, by stripping out the call options. Our analysis suggests that there may be days when the optionless municipal curve is actually inverted. This phenomenon deserves scholarly attention beyond the scope of this article.

Municipal Bonds: tax-aware strategy

OPTIMAL MUNICIPAL BOND PORTFOLIOS FOR DYNAMIC TAX MANAGEMENT

by Andrew Kalotay (2016)

- Tax-loss selling of municipal bonds is typically an ad hoc year-end exercise.
- Right to execute a tax-beneficial trade is considered to be a valuable option.
- Long-duration bonds trading at a premium are best poised to achieve superior performance; bonds purchased below par are unsuitable for tax management. The incremental return from dynamic management is significant, particularly when short-term gains are available to offset short-term losses.



Side note:
Tax aware SMA's
(direct indexing) is
a growth segment
in equities too

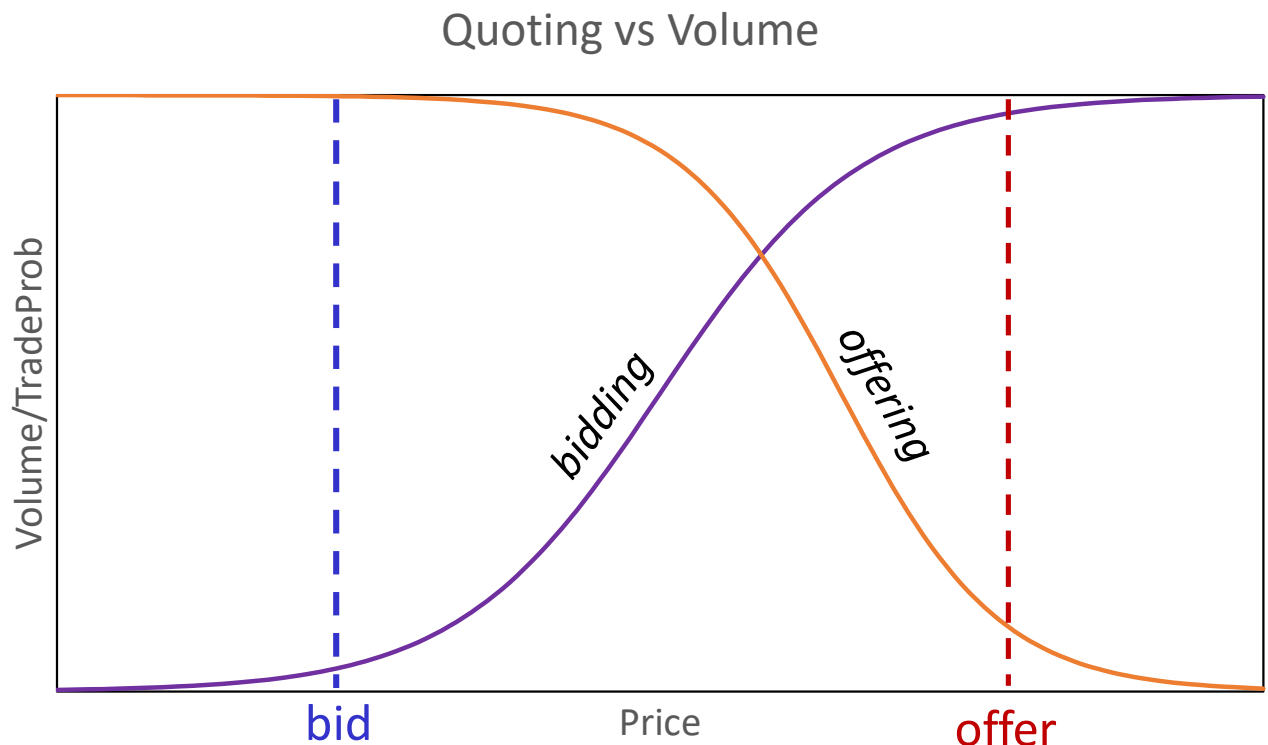
Figure 5 Performance of optionless premium and par bonds under aggressive tax management.

Municipal Bonds: systematic trading

Systematic Dealer perspective:

- buy auctioned odd-lots in competition via RFQs aka bid-wanted
- Sell these odd lots in competition via trading venue
- Profit = volume x spread

- Trades (MSRB)
 - Real time
- Bond Descriptions
 - Static
- Bid-wanted
 - Real Time
- Treasury data
 - Real time



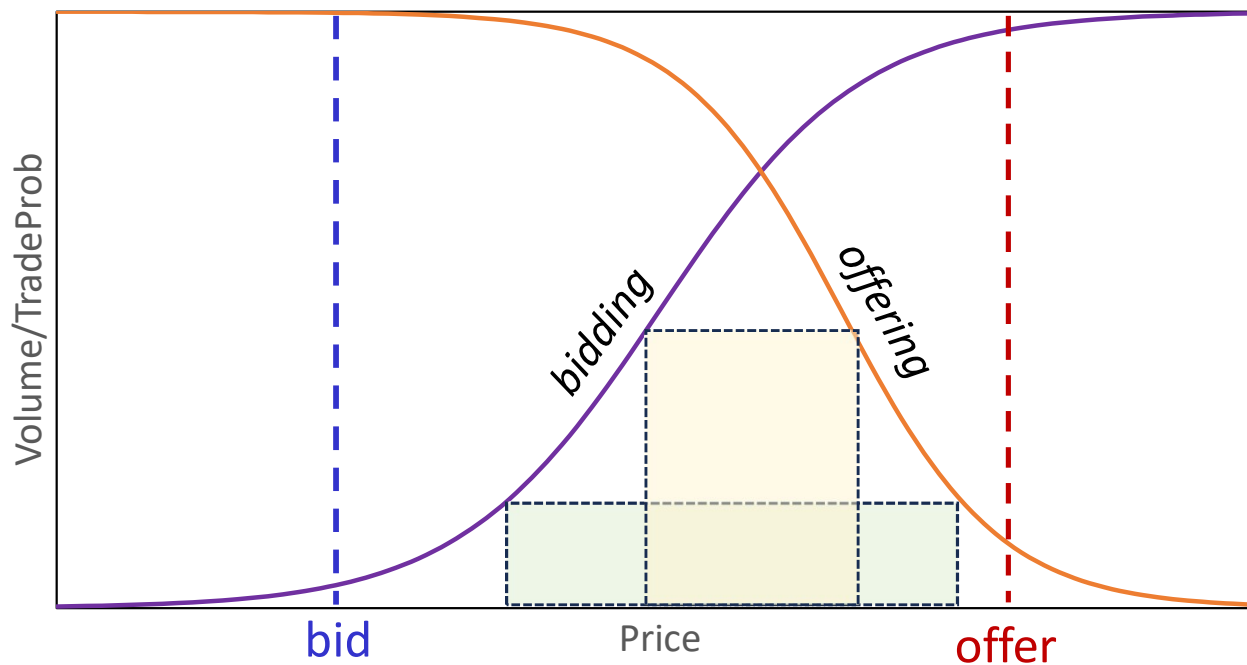
**Applicable outside of muni space*

Municipal Bonds: systematic trading

Systematic Dealer perspective:

- buy auctioned odd-lots in competition via RFQs aka bid-wanted
- Sell these odd lots in competition via trading venue
- Profit = volume x spread

Quoting vs Volume



Geometry:
Maximized area,
maximized profit

Caveats:
Bid/offer unknown
Curves unknown
Buy/sell rates unknown
Long only
Inventory cost

What would FI strat do*

Function	Broker-Dealer	Asset Manager
Models/Signals	Statistical Microstructure Low-latency Per symbol	Econometric Fundamental Alternative Data Cross-sectional
Attribution	FIFO/Trade+Hold	Brinson/Factor
Backtests	Intraday Fill Model	Daily Trade Costs and Prob.
Trading	Quoting, market-making	Execution
Risk management	Intraday, few accounts	Daily, 000's accounts
Feature	Broker-Dealer	Asset Manager
Timescale	Nanoseconds to days	Months
Developer skills emphasis	More	Less
Client facing opportunity	Less	More
Atmosphere	"Sense of urgency"	"client-oriented"
Gratification	Immediate	Delayed

**very subjective personal view*

SYSTEMATIC TRADING MTH9897

Lecture 7: Fixed Income

Quantitative Investment Framework



Misha Boroditsky, PhD
Fixed Income Analytics Lead
Bloomberg LLP
mboroditsky@hotmail.com

Teaching Assistant:
Alan Coman alancoman@gmail.com

“Classic” Equity Factors

$$R = \alpha + \beta_m \text{MKT} + \beta_s \text{SMB} + \beta_h \text{HML} + \beta_r \text{RMW} + \beta_c \text{CMA} ,$$

where

- MKT is the excess return of the market -- on the value-weighted market portfolio.
- SMB is the return on a diversified portfolio of **S**mall-cap stocks minus the return on a diversified portfolio of **B**ig-cap stocks.
- HML is the difference between the returns on diversified portfolios of stocks with **H**igh and **L**ow Book-to-Market ratios.
- RMW is the difference between the returns on diversified portfolios of stocks with **R**obust (high and steady) and **W**weak (low) profitability.
- CMA is the difference between the returns on diversified portfolios of the stocks of low and high investment firms, which we call **C**onservative and **A**ggressive. Here, low/high investment means reinvestment ratio is low/high.

Factor Investing in Credit

Equity-style factors:

- Size (SMB)
- Value
- Carry High/Low
- Quality (low volatility)
- Momentum (Jeggadish & Titman)

Credit:

Houweling and Van Zundert (2017) bond-level factors similar to Equity (dates 1994-2015) all significant:

- Size: market value of all company's bonds;
- Low-risk: Market value of short-maturity bonds with a high credit rating
- Value, measured as bonds whose credit spread was high relative to a model-implied fair spread;
- momentum, bonds with high past returns.

“Factor Investing in Credit”, Henke, Kaufmann, Messow, Fang-Klingler, 2021